

刘意 12413257

Chapter 21:

7. Solution:

To ensure the particles remaining, q_3 should be negative.

Set horizontal right as the positive direction.

$$\vec{F}_{12} = k \frac{q_1 q_2}{L^2} (-\hat{r})$$

$$\vec{F}_{13} = k \frac{q_1 |q_3|}{x^2} \hat{r}$$

$$\vec{F}_{23} = k \frac{q_2 |q_3|}{(L-x)^2} (-\hat{r})$$

$$\vec{F}_{12} + \vec{F}_{13} = \vec{0}$$

$$\vec{F}_{23} - \vec{F}_{12} = \vec{0}$$

$$\therefore x = 2\text{cm}, q_3 = -\frac{9}{16} q$$

$$(a) x = 2.00\text{cm}, (b) y = 0, (c) q_3/q = -\frac{9}{16}$$

10. Solution:

Since ions 3 and 4 are at identical angles, only thinking about x axis is enough.

Set horizontal right as the positive direction.

$$F_{21} = -k \frac{e^2}{R^2}$$

$$F_{23x} = F_{24x} = k \frac{eq}{(R/\cos\theta)^2} \cos\theta$$

$$F_{21} + F_{23x} + F_{24x} = 0$$

$$\therefore \cos\theta = \sqrt[3]{\frac{e}{2q}}, \theta \text{ decreases as } q \text{ decreases.}$$

$$(a) \text{ largest: } \theta_1 = \cos^{-1} \sqrt[3]{\frac{e}{2.5e}} = 62.3^\circ$$

$$(b) \text{ second largest: } \theta_2 = \cos^{-1} \sqrt[3]{\frac{e}{2.4e}} = 60.0^\circ$$

(c) third largest: $\theta_3 = \sqrt[3]{\frac{e}{2.3e}} = 56.6^\circ$

16. Solution:

In experiment 1,

charge operation			
	A	B	C
①	2Q	-12Q	2Q
②	2Q	-5Q	-5Q
③	-1.5Q	-5Q	-1.5Q

In experiment 2

charge operation			
	A	B	C
①	4Q	-6Q	-6Q
②	-Q	-6Q	-Q

$$\therefore F_2 / F_1 = \frac{|Q_{A2} Q_{B2}|}{|Q_{A1} Q_{B1}|} = \frac{4}{5} = 0.8$$

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Chapter 22:

10. Solution:

$$dq = \frac{dx}{L} q$$

$$d\vec{E} = k \frac{dq}{R^2 + x^2} \vec{E}_s = k \frac{dq}{R^2 + x^2} (-\sin\theta \vec{E}_x + \cos\theta \vec{E}_y)$$

$$= k \frac{dx}{L} q \left(-\frac{x}{(R^2 + x^2)^{3/2}} \vec{E}_x + \frac{R}{(R^2 + x^2)^{3/2}} \vec{E}_y \right)$$

$$\therefore \vec{E} = \int_{-\frac{L}{2}}^{\frac{L}{2}} d\vec{E} = \frac{kq}{L} \left(-\int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{x dx}{(R^2 + x^2)^{3/2}} \vec{E}_x + \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{R dx}{(R^2 + x^2)^{3/2}} \vec{E}_y \right)$$

$$= \frac{kq}{L} \cdot \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{R dx}{(R^2 + x^2)^{3/2}} \vec{E}_y$$

$$= \frac{kq}{L} \left[\frac{x}{R\sqrt{R^2 + x^2}} \right]_{-\frac{L}{2}}^{\frac{L}{2}} \vec{E}_y$$

$$= (13.875 \text{ N/C}) \vec{E}_y$$

(a) 13.875 N/C

(b) form a 90° angle with the positive direction with x.

13. Solution:

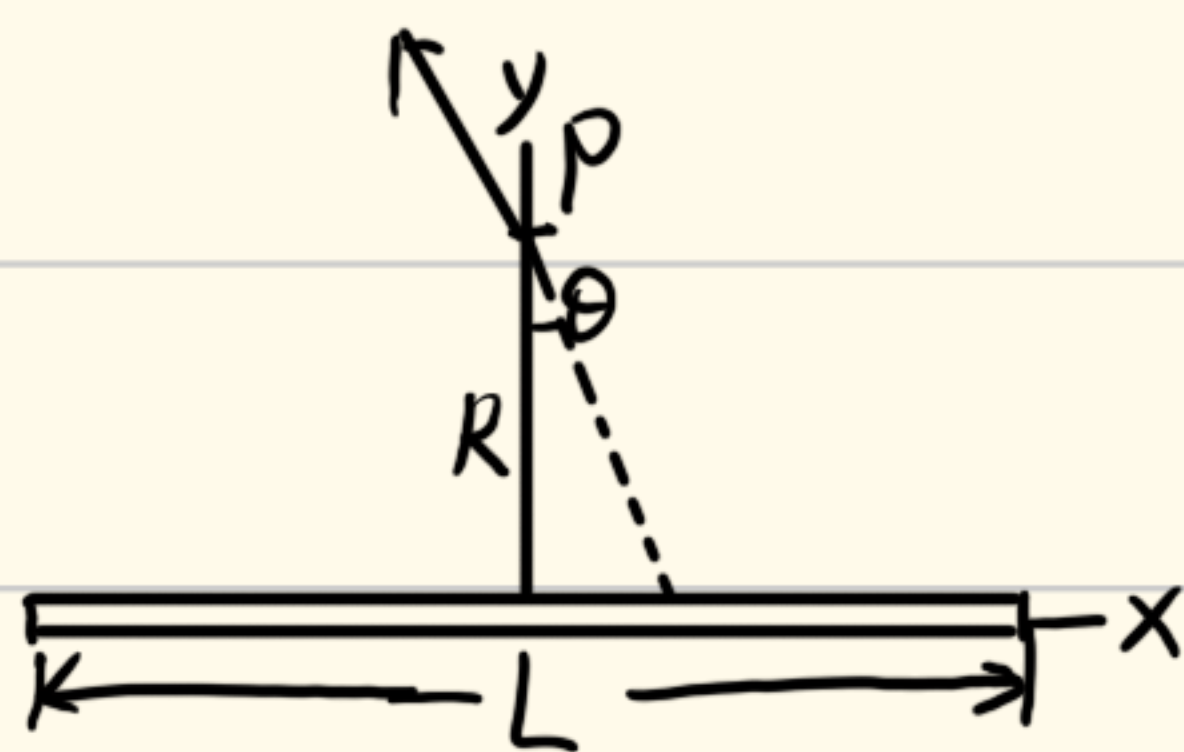
$$(a) \lambda = \frac{-q}{L} = -5.19 \times 10^{-14} \text{ C/m}$$

$$(b)(c) dq = \lambda dx = \frac{-q}{L} dx$$

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{|dq|}{(a+x)^2} \quad (x \text{ is the distance between the infinitesimal to the right end of the rod}).$$
$$= \frac{1}{4\pi\epsilon_0} \frac{q dx}{L(a+x)^2}$$

$$\therefore \vec{E} = \frac{q}{4\pi\epsilon_0 L} \int \frac{dx}{(a+x)^2} = \frac{q}{4\pi\epsilon_0 L} \left[-\frac{1}{a+x} \right]_0^L = 4.48 \times 10^{-3} \text{ N/C}$$

direction: 180° , in other words, along the negative direction of the x axis.



(d)(e) Since $a \gg L$, L can be treated as a point charge.

$$\text{So } E_3 = E_4 = \frac{1}{4\pi\epsilon_0} \frac{|q|}{a^2} = 1.52 \times 10^{-8} \text{ N/C}$$

41. Solution:

$$W = -\Delta U = -pE(\cos\theta_2 - \cos\theta_1) = 2.56 \times 10^{-23} \text{ J}$$

50. Solution:

$$\vec{v}_x = |\vec{v}_0| \cos\theta_0 \hat{i} = 3.06 \times 10^6 \text{ m/s}$$

$$\vec{v}_{y0} = |\vec{v}_0| \sin\theta_0 \hat{j} = 2.57 \times 10^6 \text{ m/s}$$

$$x = v_x t \quad \therefore t = 9.80 \times 10^{-7} \text{ s}$$

$$eE = ma \quad a = 8.79 \times 10^{11} \text{ m/s}^2$$

$$v_y = v_{y0} - at \quad v_y = 1.71 \times 10^6 \text{ m/s}$$

$$\therefore \vec{v} = \vec{v}_x + \vec{v}_y = (3.06 \times 10^6 \text{ m/s}) \hat{i} + (1.71 \times 10^6 \text{ m/s}) \hat{j}$$

54. As the graph shows, $E = 0.5 E_m$ at $z = 0.5 z_s$.

$$\sigma = \frac{Q}{\pi R^2}$$

$$z = 0: E_m = \frac{\sigma}{2\epsilon_0}$$

$$z = 0.5 z_s: 0.5 E_m = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{0.5 z_s}{\sqrt{0.25 z_s^2 + R^2}} \right)$$

$$\therefore R = 13.9 \text{ cm}$$

59. Solution:

To get a general conclusion, we use q and r to represent the charge and radius.

$$dq = \frac{2q}{\pi} d\theta$$

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \vec{e}_r = \frac{1}{4\pi\epsilon_0} \frac{2q d\theta}{\pi r^2} (\sin\theta \vec{e}_x + \cos\theta \vec{e}_y)$$

$$\begin{aligned} \therefore \vec{E} &= \frac{q}{2\pi^2\epsilon_0 r^2} \left(\int_0^{\frac{\pi}{2}} \sin\theta d\theta \vec{e}_x + \int_0^{\frac{\pi}{2}} \cos\theta d\theta \vec{e}_y \right) \\ &= \frac{q}{2\pi^2\epsilon_0 r^2} (\vec{e}_x + \vec{e}_y). \end{aligned}$$

$$\therefore \vec{E}_1 = \frac{Q}{2\pi^2\epsilon_0 R^2} (\vec{e}_x + \vec{e}_y)$$

$$\vec{E}_2 = \frac{-4Q}{2\pi^2\epsilon_0 \cdot 4R^2} (\vec{e}_x + \vec{e}_y)$$

$$\vec{E}_3 = \frac{9Q}{2\pi^2\epsilon_0 \cdot 9R^2} (\vec{e}_x + \vec{e}_y)$$

$$\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$= \frac{Q}{2\pi^2\epsilon_0 R^2} (\vec{e}_x + \vec{e}_y)$$

$$= (9.16 \times 10^6 \text{ N/C}) (\vec{e}_x + \vec{e}_y)$$

$$(a) E = \sqrt{E_x^2 + E_y^2} = 1.30 \times 10^7 \text{ N/C}$$

(b) 45° clockwise with the positive x direction.

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Chapter 23:

9. Select the S_1 surface as the G -surface.

$$Q_{\text{enc}} = q + \int_a^R \frac{A}{r} 4\pi r^2 dr = q + 2\pi A(R^2 - a^2)$$

$$\text{Area} = 4\pi R^2$$

$$\therefore E \cdot \text{Area} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$\therefore E$ in the shell is to be uniform.

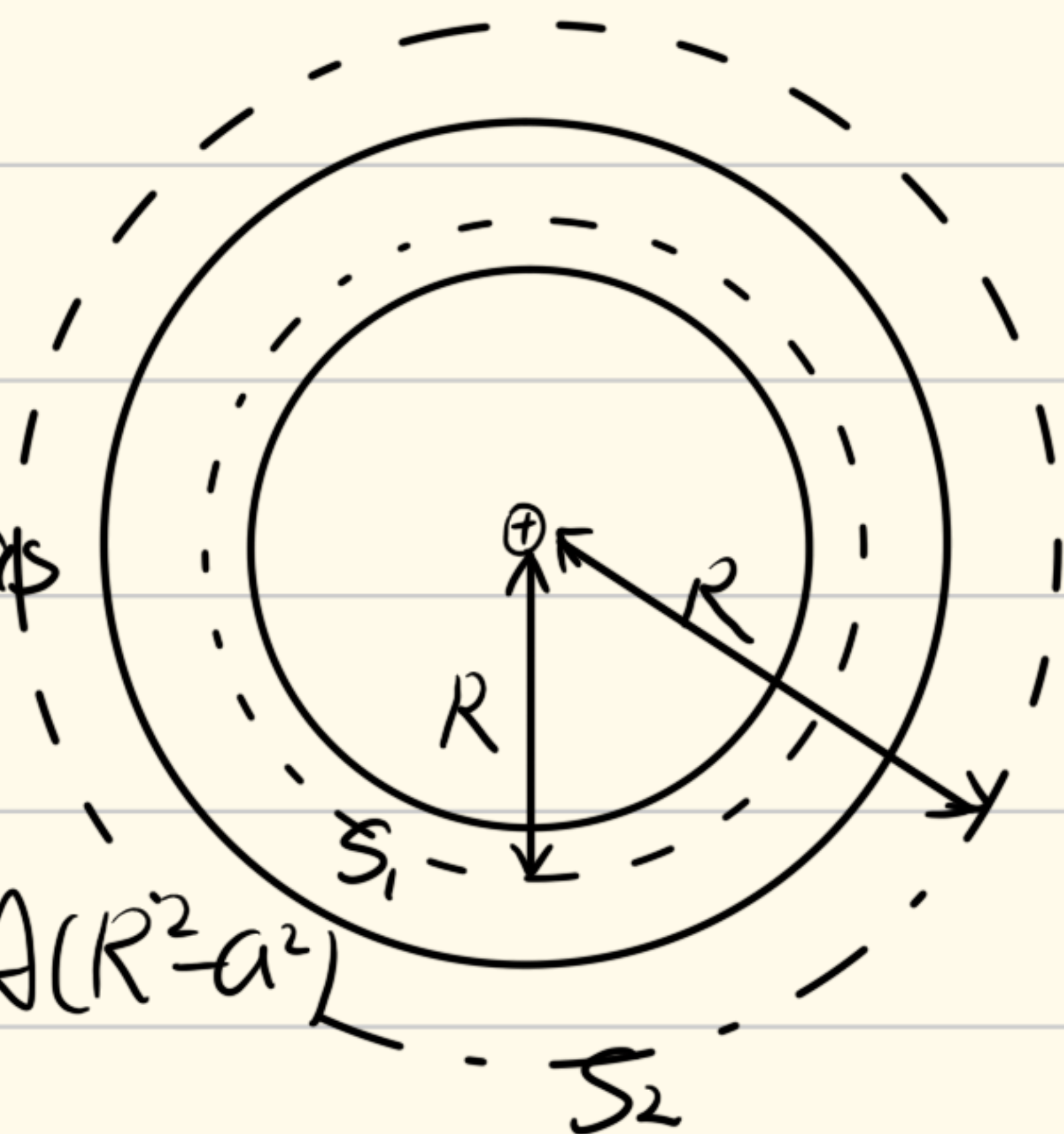
$\therefore E$ is independent of r .

$$\therefore E = \frac{Q_{\text{enc}}}{\epsilon_0 \text{Area}} = \frac{q + 2\pi A(R^2 - a^2)}{\epsilon_0 \cdot 4\pi R^2}$$

$$= \frac{A}{2\epsilon_0} + \frac{q - 2\pi Aa^2}{4\pi R^2 \epsilon_0}$$

To let E be independent of r ,

$$q - 2\pi Aa^2 = 0 \quad \therefore A = \frac{q}{2\pi a^2} = 5.3 \times 10^{-11} \text{ C/m}^2$$



$$20. \vec{E}_{\text{up}} = \frac{\sigma}{2\epsilon_0} \vec{j}$$

$$\vec{E}_{\text{down}} = -\frac{\sigma}{2\epsilon_0} \vec{j}$$

$$(a) \vec{E}_{\text{above}} = \vec{E}_{\text{up}} - \vec{E}_{\text{down}} = \frac{\sigma}{\epsilon_0} \vec{j} = (2.61 \times 10^{-11} \text{ N/C}) \vec{j}$$

$$(b) \vec{E}_{\text{between}} = -\vec{E}_{\text{up}} - \vec{E}_{\text{down}} = 0$$

$$(c) \vec{E}_{\text{below}} = -\vec{E}_{\text{up}} + \vec{E}_{\text{down}} = -\frac{\sigma}{\epsilon_0} \vec{j} = (-2.61 \times 10^{-11} \text{ N/C}) \vec{j}$$

23. Since $\vec{E}$ is uniform on x axis and z axis, these two directions $\phi_x = \phi_z = 0$.

$$\phi = \phi_y = |E_{y,y=0}|S - |E_{y,y=2}|S = -96 \text{ N} \cdot \text{m}^2/\text{C}$$

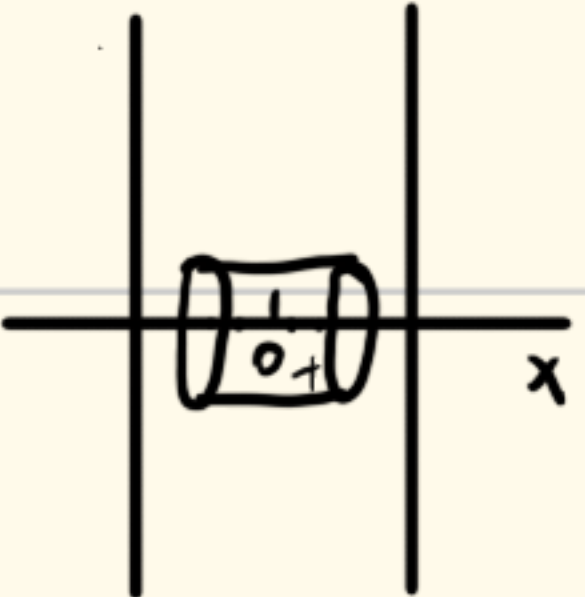
$$\therefore \phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\therefore Q_{\text{enc}} = \phi \epsilon_0 = -8.50 \times 10^{-10} \text{ C}$$

33. $E = \frac{|\sigma|}{\epsilon_0}$ (direction: towards the plate)

$$K_0 - |e| E d = 0$$

$$\therefore d = 1.45 \times 10^{-3} \text{ m}$$

47.  Select the G -surface as shown.

$$Q_{\text{enc}} = \rho \cdot \pi r^2 \cdot 2x = 2\rho \pi r^2 x$$

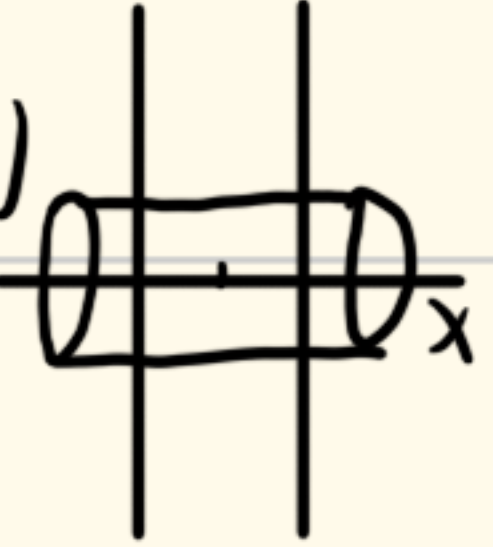
$$A = 2 \cdot \pi r^2 \quad \therefore E \cdot A = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\therefore E = \frac{\rho}{\epsilon_0} x$$

$$\therefore (a) E_1 = 0$$

$$(b) E_2 = \frac{\rho}{\epsilon_0} x_2 = 4.27 \times 10^{-7} \text{ N/C}$$

$$(c) E_3 = \frac{\rho}{\epsilon_0} x_3 = 1.00 \times 10^{-6} \text{ N/C}$$

(d)  $Q_{\text{enc}} = \rho \cdot \pi r^2 d$

$$E = \frac{Q_{\text{enc}}}{\epsilon_0 A} = \frac{\rho}{2\epsilon_0} d$$

$$\therefore E_4 = \frac{\rho}{2\epsilon_0} d = 1.00 \times 10^{-6} \text{ N/C}$$

$$54. (a) \frac{2}{5} \phi_s = \frac{q}{\epsilon_0}$$

$$\therefore q = 3.54 \times 10^{-6} \text{ C}$$

$$(b) -\frac{4}{5} \phi_s = \frac{q + q_A}{\epsilon_0} \quad \leftarrow \begin{array}{l} \text{As the Shell Theorem 1 shows,} \\ \text{shells can be regarded as point charges outside.} \end{array}$$

$$\therefore q_A = -1.06 \times 10^{-5} \text{ C}$$

$$(c) \frac{6}{5} \phi_s = \frac{q + q_A + q_B}{\epsilon_0}$$

$$\therefore q_B = 1.77 \times 10^{-5} \text{ C}$$

55. (a)(b) Select a cylinder surface as the G_1 -surface.

Φ in both the top and bottom is 0.

$$A = 2\pi \cdot 2R_2 L = 4\pi R_2 L$$

$$Q_{\text{enc}} = Q_1 + Q_2 = -Q_1$$

$$\therefore E = \frac{Q_{\text{enc}}}{\epsilon_0 A} = 3.28 \times 10^{-2} \text{ N/C}$$

$$(a) E = 3.28 \times 10^{-2} \text{ N/C} \quad (b) \text{ radially outward}$$

$$(c)(d) A = 2\pi \cdot 5R_1 L$$

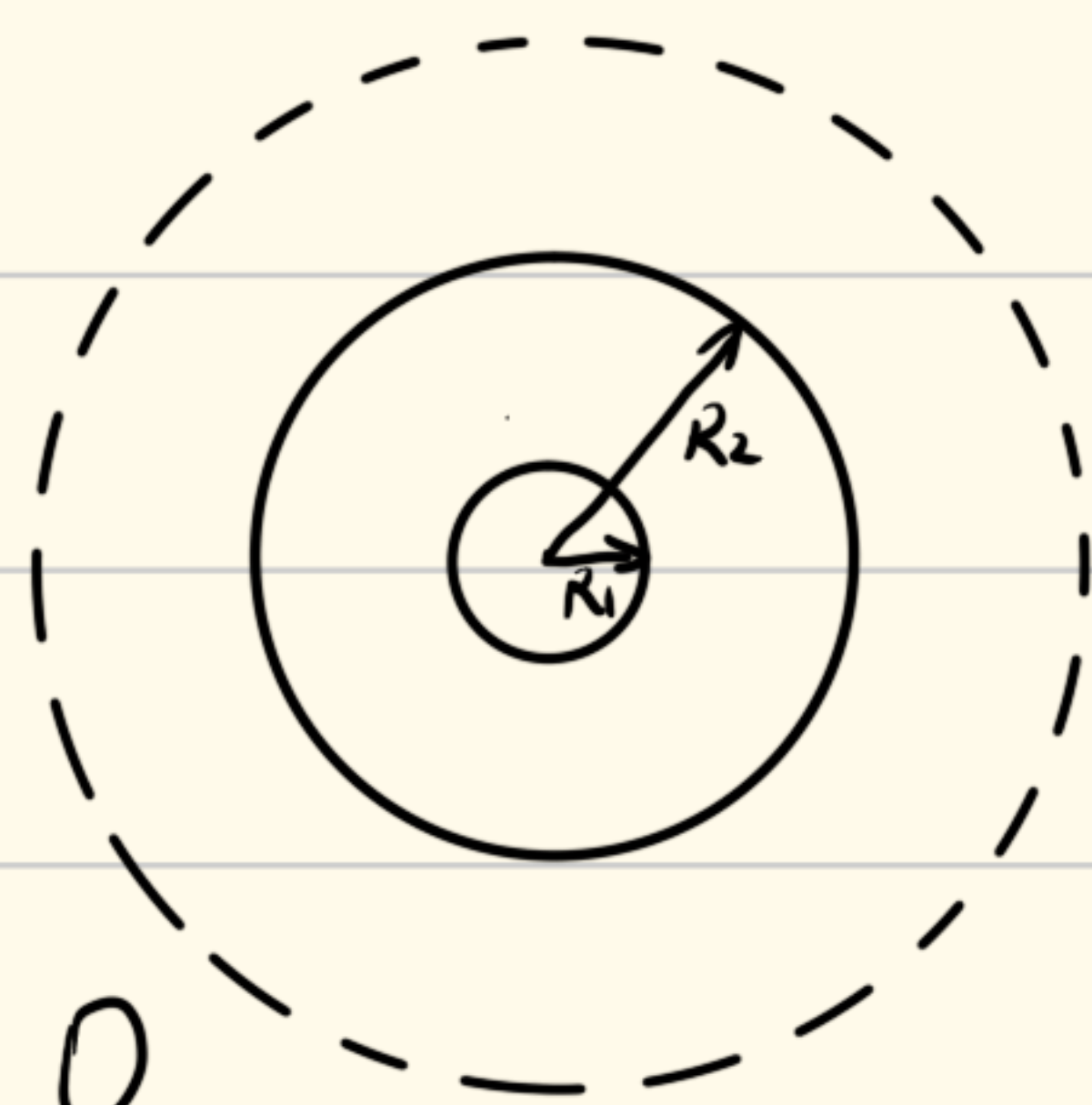
$$Q_{\text{enc}} = Q_1$$

$$\therefore E = \frac{Q_{\text{enc}}}{\epsilon_0 A} = -0.131 \text{ N/C}$$

$$(c) E = 0.131 \text{ N/C} \quad (b) \text{ radially inward}$$

$$(e) Q_{\text{enc}} = Q_1 + Q_{\text{int}} = 0 \quad \therefore Q_{\text{int}} = 5.22 \times 10^{-13} \text{ C}$$

$$(f) Q_{\text{ext}} = Q_2 - Q_{\text{enc}} = 5.22 \times 10^{-13} \text{ C}$$



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Chapter 24:

$$17. E_x = -\frac{\partial V}{\partial x} = -2.00yz^2$$

$$E_y = -\frac{\partial V}{\partial y} = -2.00xz^2$$

$$E_z = -\frac{\partial V}{\partial z} = -4.00xyz.$$

$$\therefore \vec{E}_x = (64.0 \text{ V}) \hat{i}$$

$$\vec{E}_y = (32.0 \text{ V}) \hat{j}$$

$$\vec{E}_z = (-32.0 \text{ V}) \hat{k}$$

$$\therefore \vec{E} = \vec{E}_x + \vec{E}_y + \vec{E}_z = (64.0 \hat{i} + 32.0 \hat{j} - 32.0 \hat{k}) \text{ N/C}$$

$$E = \sqrt{E_x^2 + E_y^2 + E_z^2} = 78.4 \text{ N/C}$$

$$31. V_1 = \frac{q}{4\pi\epsilon_0 d}$$

$$V_2 = \frac{-q}{4\pi\epsilon_0 d}$$

$$V_3 = \frac{-q}{4\pi\epsilon_0 (2d)}$$

$$\therefore V = 2V_1 + V_2 + V_3 = \frac{q}{4\pi\epsilon_0 d} (2 - 1 - \frac{1}{2}) = 2.11 \times 10^{-3} \text{ V}$$

$$37. (a) U = \frac{q^2}{4\pi\epsilon_0 d} = 7.49 \times 10^{-2} \text{ J}$$

$$(b) F = \frac{q^2}{4\pi\epsilon_0 d^2} = 2.497 \times 10^{-2} \text{ N}$$

$$a_A = \frac{F}{m_A} = 4.99 \text{ m/s}^2$$

$$a_B = \frac{F}{m_B} = 2.50 \text{ m/s}^2$$

$$(c) K_A + K_B - U = 0$$

$$K_A = \frac{1}{2} m_A V_A^2 \quad K_B = \frac{1}{2} m_B V_B^2$$

$$m_A V_A - m_B V_B = 0$$

$$\therefore V_A = 4.47 \text{ m/s}, V_B = 2.23 \text{ m/s}.$$

$$49. (a) V_1 = \frac{q_1}{4\pi\epsilon_0 \frac{d}{2}} = 89.9 \text{ V}$$

$$V_2 = \frac{q_2}{4\pi\epsilon_0 \frac{d}{2}} = -719 \text{ V}$$

$$\therefore V_{\frac{d}{2}} = V_1 + V_2 = -629 \text{ V}$$

(b) For sphere 1.

$$V_{11} = \frac{q_1}{4\pi\epsilon_0 R_1} = 2995.1 \text{ V}$$

$$V_{21} = \frac{q_2}{4\pi\epsilon_0 d} = -359.3 \text{ V}$$

$$\therefore V_1 = V_{11} + V_{21} = 2634 \text{ V}$$

For sphere 2,

$$V_{22} = \frac{q_2}{4\pi\epsilon_0 R_2} = -23966.8 \text{ V}$$

$$V_{12} = \frac{q_1}{4\pi\epsilon_0 d} = 44.9 \text{ V}$$

$$\therefore V_2 = V_{22} + V_{12} = -23952 \text{ V}$$

$$65. (a) E = \frac{\sigma}{2\epsilon_0} = 0.3275 \text{ N/C}$$

$$W = qEd = 3.22 \times 10^{-21} \text{ J}$$

$$(b) \Delta U = -W = qV$$

$$V = V_0 + \frac{\Delta U}{q} = -2.51 \times 10^{-2} \text{ V}$$

$$67. dq = \frac{1}{4} \cdot 2\pi r \cdot \sigma dr = \frac{1}{2} \sigma r dr$$

$$dV = \frac{dq}{4\pi\epsilon_0 s}, \quad s = \sqrt{D^2 + r^2}$$

$$\begin{aligned} \therefore V &= \int_0^R \frac{dq}{4\pi\epsilon_0 \sqrt{D^2 + r^2}} = \int_0^R \frac{\sigma r}{8\epsilon_0 \sqrt{D^2 + r^2}} dr \\ &= \frac{\sigma}{8\epsilon_0} (\sqrt{D^2 + R^2} - D) \\ &= 3.63 \times 10^{-5} \text{ V} \end{aligned}$$

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Chapter 25:

$$6. (a) q_3 = C_3 V = 2.00 \times 10^{-4} \text{ C}$$

$$(b) V_3 = V = 100 \text{ V}$$

$$(c) U_3 = \frac{1}{2} C_3 V_3^2 = 0.0100 \text{ J}$$

$$(d) \sim (i) q_1 = q_2$$

$$\therefore C_1 V_1 = C_2 V_2$$

$$V_1 + V_2 = V$$

$$\therefore V_1 = 33.3 \text{ V}, V_2 = 66.7 \text{ V}$$

$$q_1 = q_2 = C_1 V_1 = 3.33 \times 10^{-4} \text{ C}$$

$$U_1 = \frac{1}{2} C_1 V_1^2 = 5.56 \times 10^{-3} \text{ J}$$

$$U_2 = \frac{1}{2} C_2 V_2^2 = 1.11 \times 10^{-2} \text{ J}$$

$$(d) q_1 = 3.33 \times 10^{-4} \text{ C}$$

$$(e) V_1 = 33.3 \text{ V}$$

$$(f) U_1 = 5.56 \times 10^{-3} \text{ J}$$

$$(g) q_2 = 3.33 \times 10^{-4} \text{ C}$$

$$(h) V_2 = 66.7 \text{ V}$$

$$(i) U_2 = 1.11 \times 10^{-2} \text{ J}$$

$$14. C_1 = K C_{1\text{vac}} = K \frac{\epsilon_0 A}{d} = 6.64 \times 10^{-11} \text{ F}$$

$$C_2 = \frac{\epsilon_0 A}{d} = 2.21 \times 10^{-11} \text{ F}$$

$$C = C_1 + C_2 = 8.85 \times 10^{-11} \text{ F} \quad (\text{capacitors in parallel})$$

$$Q = CV = 8.85 \times 10^{-10} \text{ C}$$

$$17. (a) C_i = \frac{\epsilon_0 A}{d_i} = 9.408 \times 10^{-13} \text{ F}$$

$$q = C_i V_i = 1.505 \times 10^{-11} \text{ C}$$

$$C_f = \frac{\epsilon_0 A}{d_f} = 2.509 \times 10^{-12} \text{ F}$$

$$V_f = \frac{q}{C_f} = 6.00 \text{ V}$$

$$(b) U_i = \frac{1}{2} C_i V_i^2 = 1.20 \times 10^{-10} \text{ J}$$

$$(c) U_f = \frac{1}{2} C_f V_f^2 = 4.52 \times 10^{-11} \text{ J}$$

$$(d) W_{app} = U_f - U_i = -7.48 \times 10^{-11} \text{ J}$$

$$34. (a) \therefore C = \frac{\epsilon_0 A}{d}$$

$$\therefore \frac{C_f}{C_i} = \frac{d}{(1.50\%)d} = 2.$$

$$\therefore C_f = 2C_i = 16 \mu\text{F}$$

$$Q_{add} = C_f V - C_i V = 8.0 \times 10^{-5} \text{ C}$$

(b) Since V is constant, q in other capacitor doesn't change.

$$\therefore Q = Q_{add} = 8.0 \times 10^{-5} \text{ C}$$

$$36. C_1 = \frac{2\pi\epsilon_0 L_1}{\ln \frac{b_1}{a_1}} = 1.728 \times 10^{-12} \text{ F}$$

$$C_2 = \frac{2\pi\epsilon_0 L_2}{\ln \frac{b_2}{a_2}} = 3.612 \times 10^{-12} \text{ F}$$

$$C_2' = \frac{2\pi\epsilon_0 L_2}{\ln \frac{b_2'}{a_2}} = 2.174 \times 10^{-12} \text{ F}$$

$$C_i = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} = 1.169 \times 10^{-12} \text{ F}$$

$$C_f = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2'}} = 9.628 \times 10^{-13} \text{ F}$$

$$(a) \Delta q = C_f V - C_i V$$

$$n = \frac{\Delta q}{e}$$

$$\therefore n = 1.30 \times 10^7$$

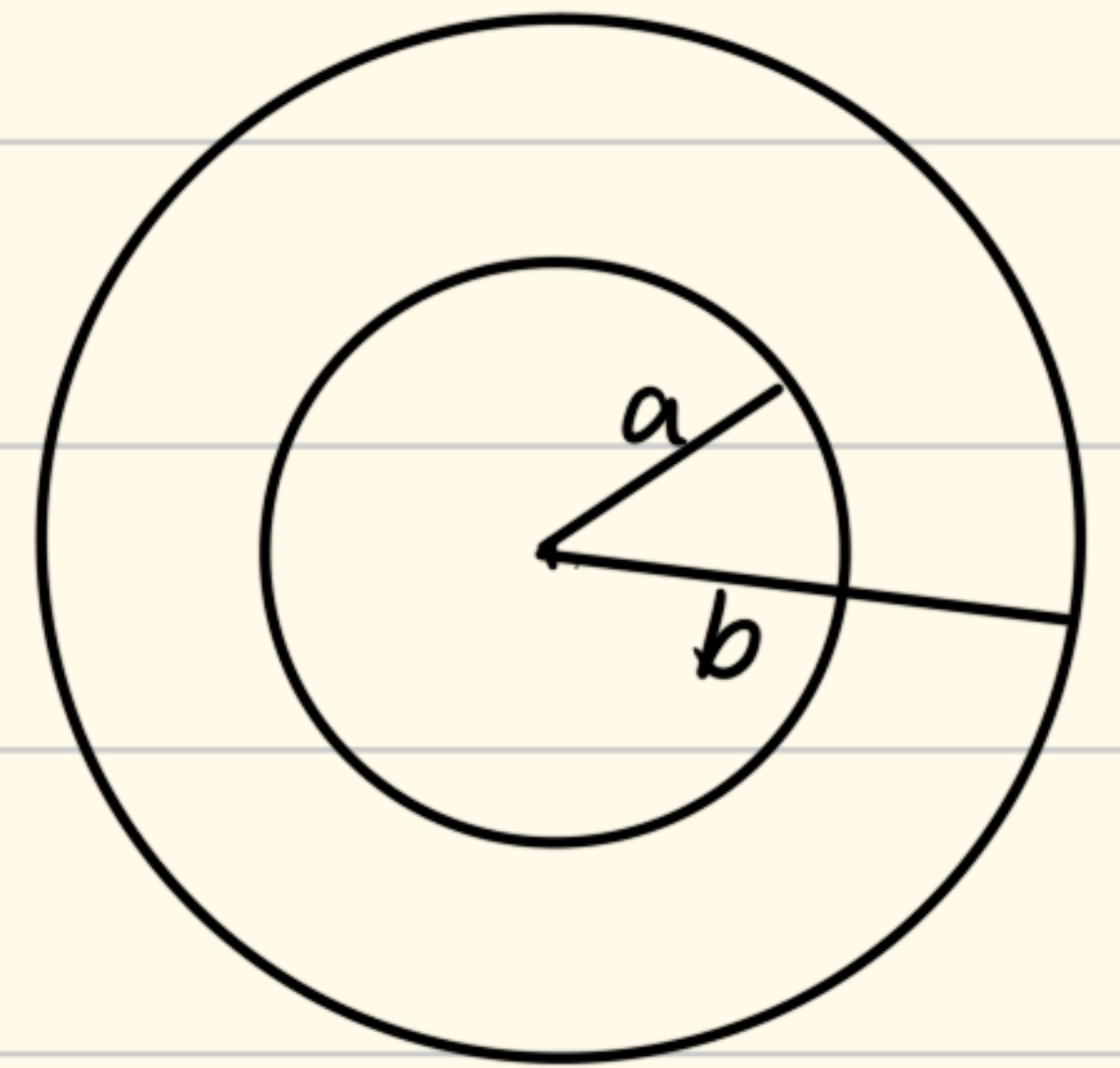
(b) Since $C_f < C_i$, $\Delta q < 0$,

electrons move away from the battery.

$$49. (a) C = k \cdot 4\pi\epsilon_0 \frac{ab}{b-a} = 3.137 \times 10^{-11} \text{ F}$$

$$(b) q = CV = 2.29 \times 10^{-9} \text{ C}$$

$$(c) q' = q(1 - \frac{1}{k}) = 1.96 \times 10^{-9} \text{ C}$$



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Chapter 26:

$$5. R = \int_0^L \frac{\rho}{A} dl$$

$$r = a + \frac{b-a}{L} l$$

$$A = \pi r^2$$

$$\therefore R = \int_0^L \frac{\rho}{\pi \left(a + \frac{b-a}{L} l\right)^2} dl$$

$$= \frac{\rho}{\pi} \left[\frac{1}{\frac{a-b}{L} \left(a + \frac{b-a}{L} l\right)} \right]_0^L = \frac{\rho L}{(a-b)\pi} \cdot \frac{a-b}{ab} = \frac{\rho L}{\pi ab}$$

$$= 2.08 \times 10^6 \Omega$$

$$24. \Delta V = - \int_0^{D+\Delta r} E dr = - \int_0^{D+\Delta r} \rho \frac{I_0}{2\pi r^2} dr$$

$$= \frac{\rho I_0}{2\pi} \int_0^{D+\Delta r} \left(-\frac{1}{r^2}\right) dr$$

$$= \left[\frac{\rho I_0}{2\pi} \cdot \frac{1}{r} \right]_0^{D+\Delta r} = -177.3 V$$

$$I = \frac{|kV|}{R} = 4.4 \times 10^{-2} A$$

$$25. R_c = \rho_c \frac{L_c}{A_c} = 0.6366 \Omega$$

$$R_d = \rho_d \frac{L_d}{A_d} = 1.2732 \Omega$$

$$(a) V_{12} = I R_c = 1.27 V$$

$$(b) V_{23} = I R_d = 2.55 V$$

$$(c) P_1 = I^2 R_c = 2.55 W$$

$$(d) P_2 = I^2 R_d = 5.09 W$$

$$27. (a) I = \frac{V}{R} = 3.83 \times 10^{-2} \text{ A}$$

$$(b) J = \frac{I}{A} = 142 \text{ A/m}^2$$

$$(c) J = neV_d$$

$$\therefore V_d = 1.66 \times 10^{-2} \text{ m/s}$$

$$(d) E = \frac{V}{L} = 306 \text{ V/m}$$



$$R = 935 \Omega$$

$$n = 5.33 \times 10^{22} / \text{m}^3$$

$$45. (a) dA = 2\pi r dr$$



$$I = \int_0^R J_a dA = \int_0^R \frac{J_0 r}{R} 2\pi r dr = \frac{2\pi J_0}{R} \int_0^R r^2 dr$$

$$= \frac{2\pi J_0}{R} \cdot \frac{R^3}{3} = \frac{2\pi J_0 R^2}{3} = 0.821 \text{ A}$$

$$(b) dA = 2\pi r dr$$

$$I = \int_0^R J_b dA = \int_0^R J_0 \left(1 - \frac{r}{R}\right) 2\pi r dr$$

$$= \frac{2\pi J_0}{R} \int_0^R (Rr - r^2) dr$$

$$= \frac{2\pi J_0}{R} \cdot \frac{1}{6} R^3 = \frac{\pi J_0 R^2}{3} = 0.411 \text{ A}$$

$$(c) (a), J_a = J_0 r / R.$$

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Chapter 27:

$$28, \mathcal{E} = V_1 + I_1 r = I_1 (R_1 + r)$$

$$\mathcal{E} = V_2 + I_2 r = I_2 (R_2 + r)$$

$$(a) R = 6.67 \times 10^2 \Omega \quad (b) \mathcal{E} = 0.233 V$$

$$(c) P_e = \frac{V_2^2}{R_2} = 1.875 \times 10^{-5} W$$

$$P_{light} = 1.0 \times 10^{-2} W$$

$$\eta = \frac{P_e}{P_{light}} \times 100\% = 0.19\%$$

37, Assume the current direction in battery 1 is down.

$$-\mathcal{E}_1 - I_1 R_1 + \mathcal{E}_2 - I_2 R_2 - I_1 R_1 = 0$$

$$-\mathcal{E}_1 - I_1 R_1 - (I_1 - I_2) R_1 + \mathcal{E}_3 - (I_1 - I_2) R_1 - I_1 R_1 = 0$$

$$\therefore I_1 = 0.67 A, \quad I_2 = 0.33 A.$$

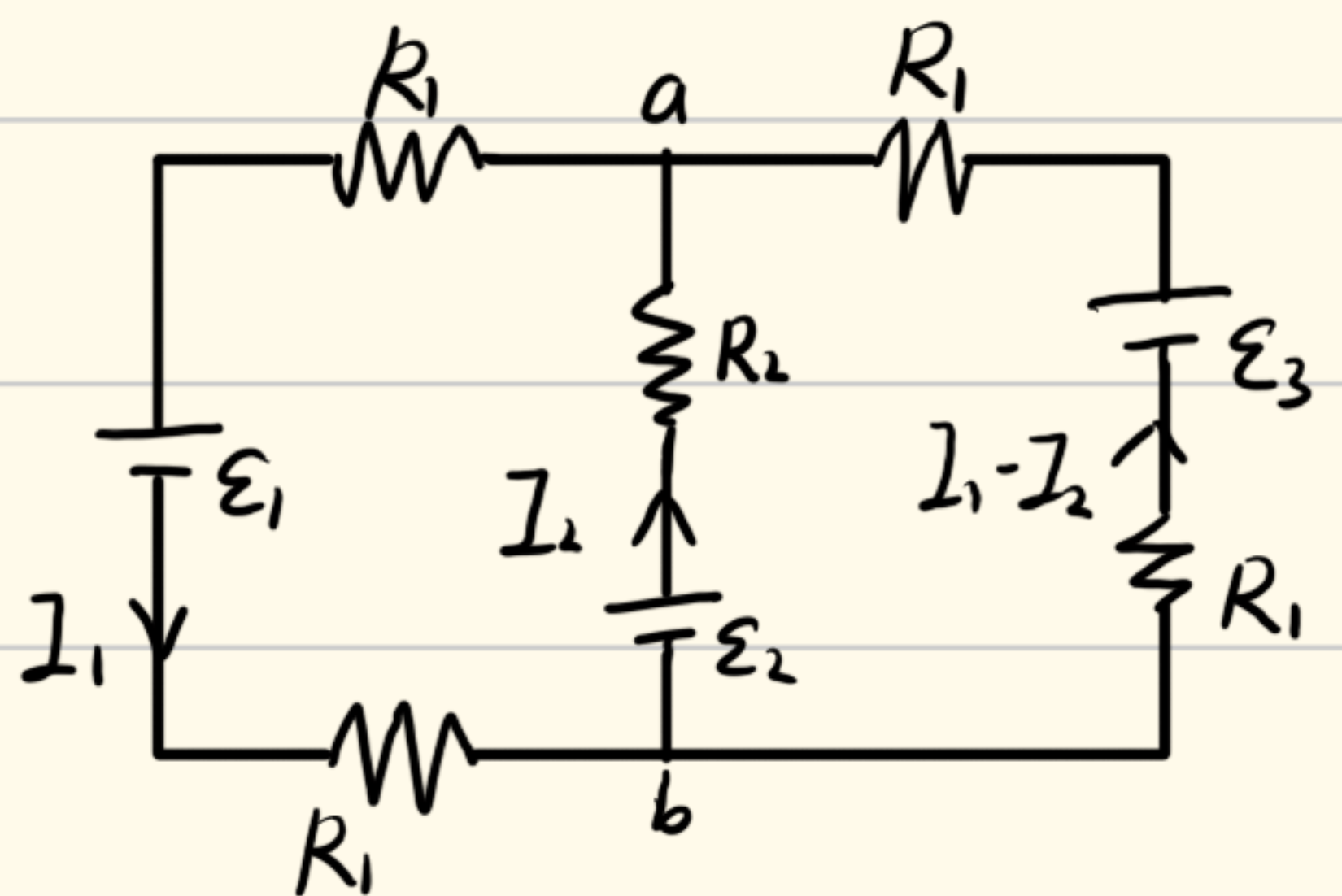
$$(a) I_1 = 0.67 A \quad (b) \text{down}$$

$$(c) I_2 = 0.33 A \quad (d) \text{up}$$

(e) absorbing

$$(f) P_{abs} = I_1 \mathcal{E}_1 = 1.34 W$$

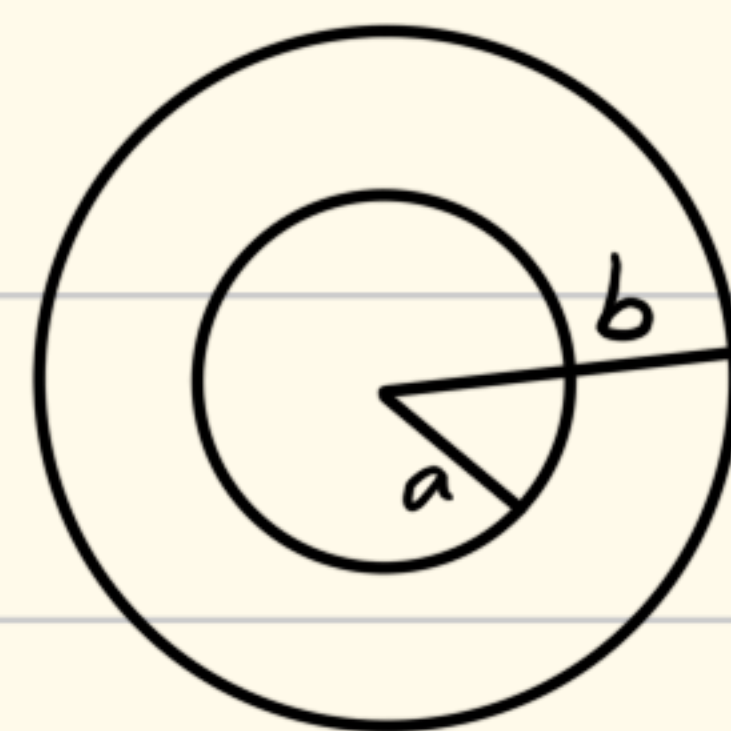
$$(g) V_a - V_b = \mathcal{E}_2 - I_2 R_2 = 3.34 V$$



$$41. (a) R_{Cu} = \rho_{Cu} \frac{l}{\pi a^2} \quad R_{Al} = \rho_{Al} \frac{l}{\pi (b^2 - a^2)}$$

$$i_1 = \frac{U}{R_{Cu}} \quad i_2 = \frac{U}{R_{Al}} \quad \therefore \frac{i_1}{i_2} = \frac{a^2 \rho_{Al}}{(b^2 - a^2) \rho_{Cu}} =$$

$$i = i_1 + i_2$$



$$(a) i_1 = 0.842 A \quad (b) i_2 = 1.16 A$$

$$(c) R = \frac{V}{i} = 6.00 \Omega$$

$$R = R_1 + R_2 = \left(\frac{\rho_{Cu}}{\pi a^2} + \frac{\rho_{Al}}{\pi (b^2 - a^2)} \right) l$$

$$\therefore l = 166 m$$

$$47. (a) i_1 = \left(\frac{\mathcal{E}}{R} \right) e^0$$

$$R = R_1 + \frac{1}{\frac{1}{R_2} + \frac{1}{R_3}} = 1.095 M\Omega$$

$$\therefore i_1 = 1.1 mA$$

$$(b) V_1 = i_1 R_1 = 800 V$$

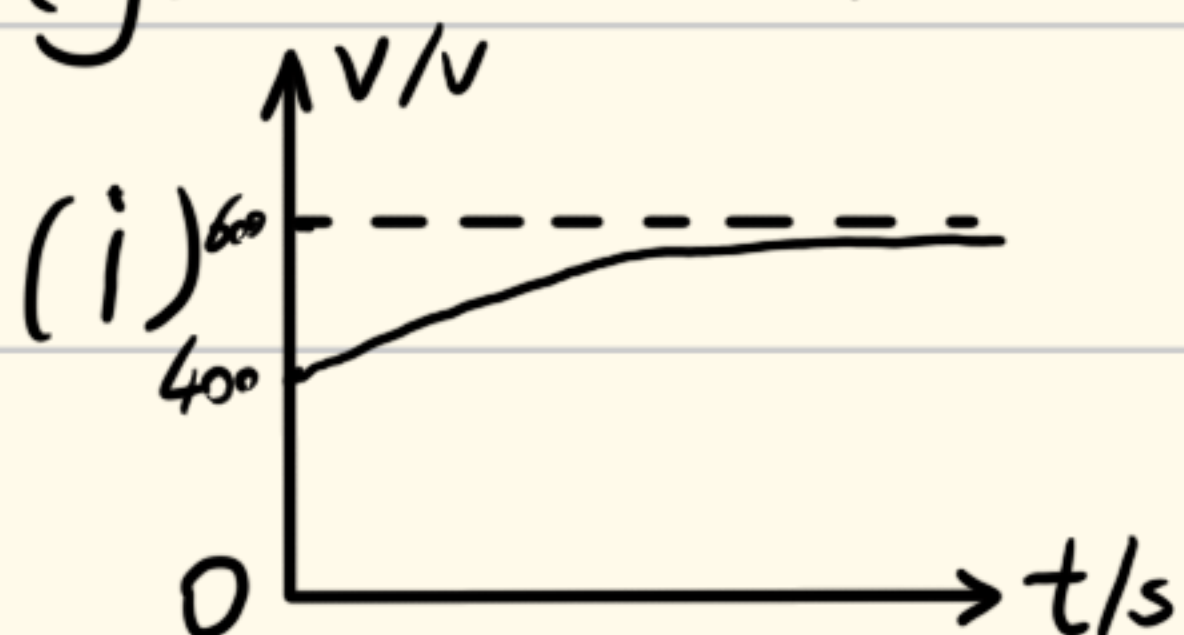
$$i_2 = \frac{\mathcal{E} - V_1}{R_2} = 0.55 mA$$

$$(c) i_3 = i_1 - i_2 = 0.55 mA$$

$$(d) (e) i_1 = i_2 = \frac{\mathcal{E}}{R_1 + R_2} = 0.82 mA$$

$$(f) i_3 = 0$$

$$(g) V_2 = \mathcal{E} - V_1 = 400 V \quad (h) V_2 = \frac{R_2}{R_1 + R_2} \mathcal{E} = 600 V$$



$$62. (a) U = \frac{q^2}{2C}$$

$$\therefore q_0 = 1.1 \times 10^{-3} \text{ C}$$

$$(b) i_0 = -\frac{q_0}{RC} e^0 = -1.1 \times 10^{-3} \text{ A}$$

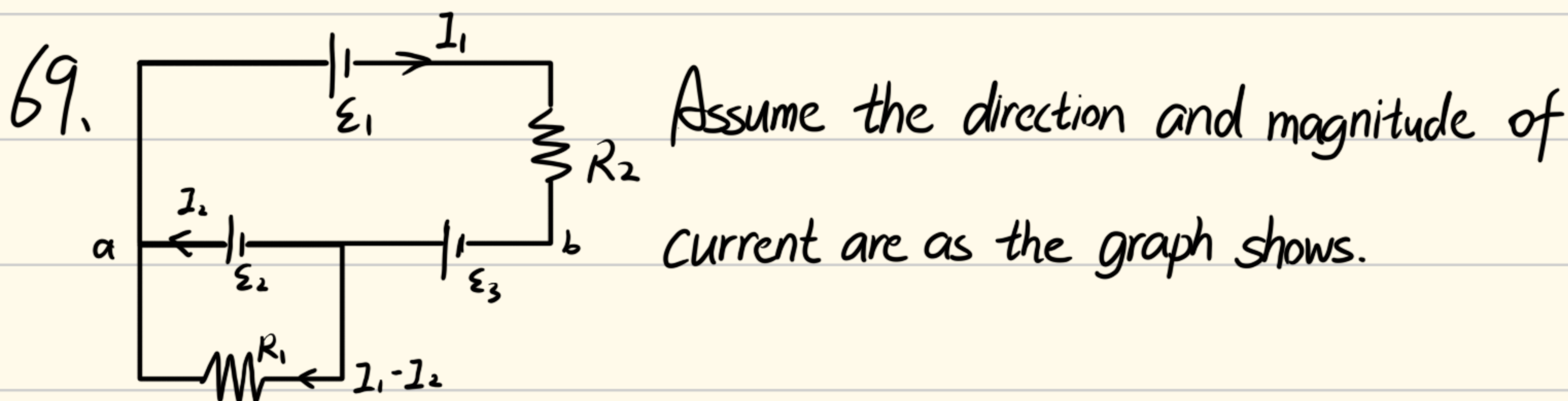
$$(c) \therefore q = q_0 e^{-\frac{t}{RC}}$$

$$\therefore V_C = \frac{q}{C} = 1.1 \times 10^3 e^{-t} \text{ V}$$

$$(d) \therefore i = \frac{dq}{dt} = -\frac{q_0}{RC} e^{-\frac{t}{RC}}$$

$$\therefore V_R = |iR| = 1.1 \times 10^3 e^{-t} \text{ V}$$

$$(e) P_{\text{thc}} = i^2 R = 1.21 e^{-2t} \text{ W}$$



$$-\varepsilon_1 - I_1 R_2 + \varepsilon_3 + \varepsilon_2 = 0$$

$$-\varepsilon_1 - I_1 R_2 + \varepsilon_3 - (I_1 - I_2) R_1 = 0$$

$$\therefore I_1 = 0.16 \text{ A}, \quad I_2 = 0.06 \text{ A}.$$

$$(a) I_1 - I_2 = 0.10 \text{ A}.$$

$$(b) I_1 = 0.16 \text{ A}$$

$$(c) V_{ab} = \varepsilon_2 + \varepsilon_3 = 14 \text{ V}$$

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Chapter 28:

$$10. \vec{L}_1 = (1.0\text{m})\hat{i} + (\sqrt{3}\text{m})\hat{j}$$

$$\vec{L}_2 = (1.0\text{m})\hat{i} + (-\sqrt{3}\text{m})\hat{j}$$

$$(a) \vec{F}_1 = i \vec{L}_1 \times \vec{B}_1 + i \vec{L}_2 \times \vec{B}_1 = (-28\hat{j})\text{N}$$

$$(b) \vec{F}_2 = i \vec{L}_1 \times \vec{B}_2 + i \vec{L}_2 \times \vec{B}_2 = \vec{0}$$

$$15. (a) K = \frac{1}{2}mv^2$$

$$F = eVB = \frac{mv^2}{r}$$

$$f = \frac{v}{2\pi r}$$

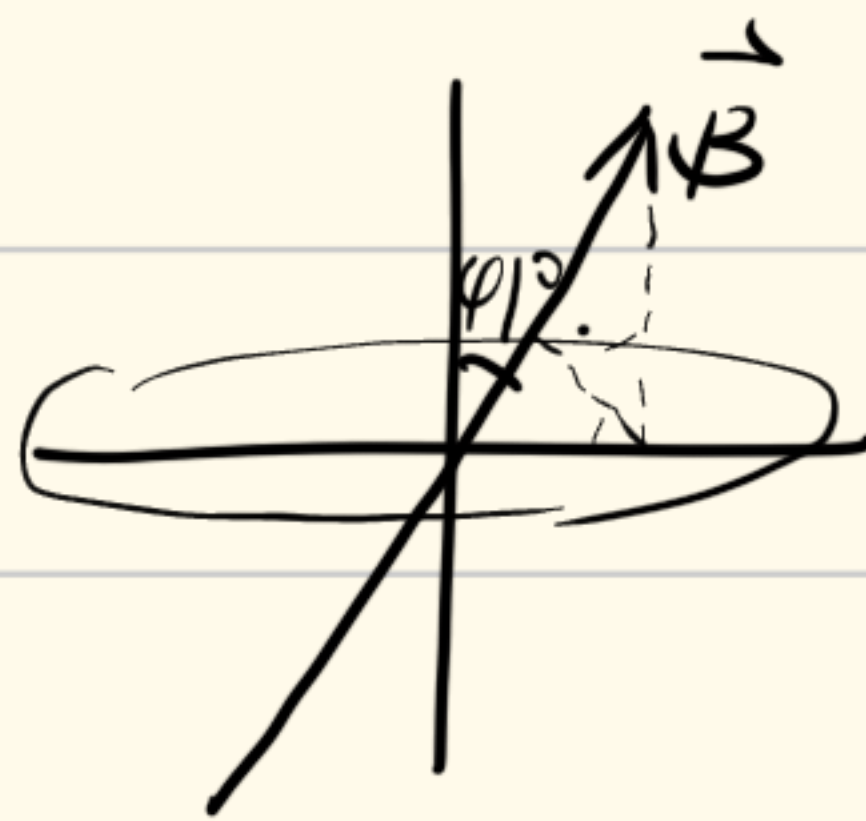
$$\therefore f = 1.96 \times 10^6 \text{Hz}$$

$$(b) r = 0.662\text{m}.$$

$$\begin{aligned} 20. (a) \vec{\mu} &= A i \hat{A} \\ &= \pi r^2 i \hat{A} \\ &= (0.226 \text{Am}^2) \hat{A}. \end{aligned}$$

$$\therefore \mu = 0.226 \text{Am}^2$$

$$(b) \tau = \vec{\mu} \times \vec{B} = \mu B \sin 41^\circ = 1.78 \text{J}$$



$$\begin{aligned}
 30. \vec{F} &= \int_{x_1}^{x_2} i d(\vec{L} \times \vec{B}) \\
 &= i \int_{x_1}^{x_2} 8x^2 dx \cdot \hat{k} \times 10^{-3} \\
 &= i \times 10^{-3} \hat{k} \left[\frac{8}{3} x^3 \right]_1^3 = (-0.485 \text{ N}) \hat{k} \\
 \therefore \vec{F} &= (-0.485 \text{ N}) \hat{k}
 \end{aligned}$$

$$\begin{aligned}
 46. (a) U_i &= K + U_f \\
 U_i &= -\mu B \cos \theta_1 \\
 U_f &= -\mu B \cos \theta \\
 \therefore \theta_1 &= 82.5^\circ
 \end{aligned}$$

$$\begin{aligned}
 (b) \text{According to the conservation of energy,} \\
 \theta_2 &= -\theta_1 = -82.5^\circ
 \end{aligned}$$

$$\begin{aligned}
 47. n &= \frac{Bi}{eVl} \\
 \therefore V &= \frac{Bi}{enl} = 3.64 \times 10^{-5} \text{ V}
 \end{aligned}$$

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Chapter 29:

$$7. B = \frac{\mu_0 i}{2\pi a} = 1.11 \times 10^{-5} \text{ T}$$

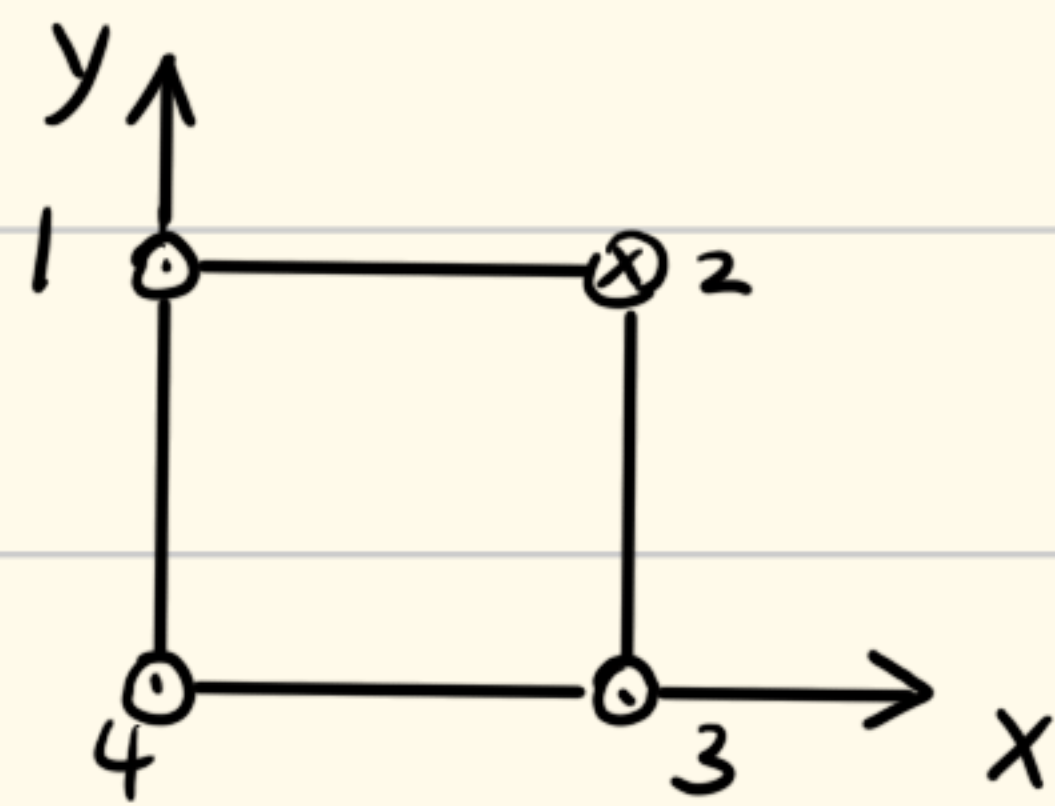
$$\vec{F}_{41}/L = (iLB)\hat{j} = (8.33 \times 10^{-5} \text{ N/m})\hat{j}$$

$$\vec{F}_{43}/L = (iLB)\hat{j} = (8.33 \times 10^{-5} \text{ N/m})\hat{j}$$

$$F_{42}/L = iL \frac{B}{\sqrt{2}} = (5.89 \times 10^{-5} \text{ N/m})\hat{j}$$

$$\therefore \vec{F}_{42}/L = (-4.17 \times 10^{-5} \text{ N/m})\hat{i} + (-4.17 \times 10^{-5} \text{ N/m})\hat{j}$$

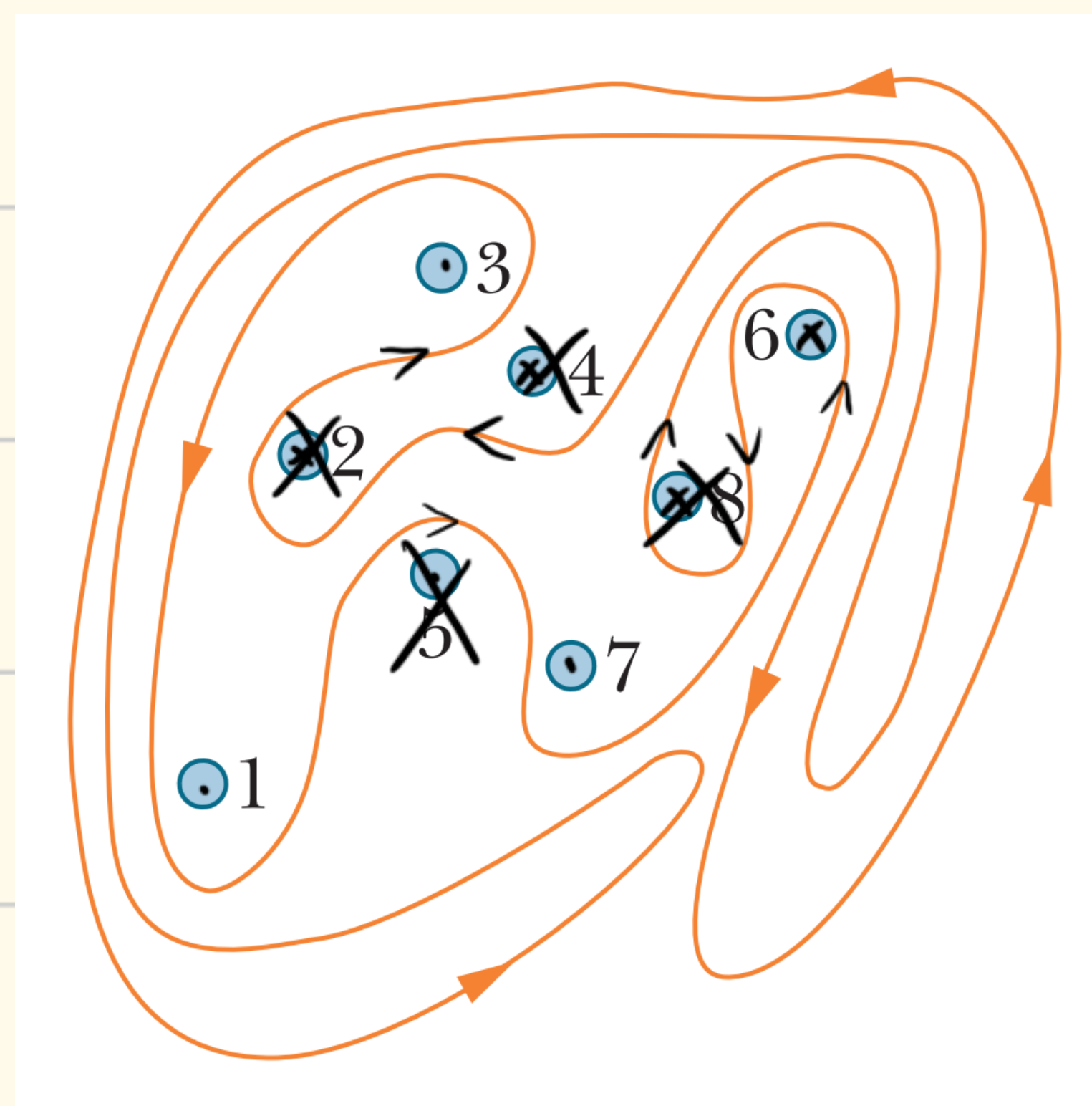
$$\vec{F}/L = \vec{F}_{41}/L + \vec{F}_{42}/L + \vec{F}_{43}/L = (4.16 \times 10^{-5} \text{ N/m})\hat{i} + (4.16 \times 10^{-5} \text{ N/m})\hat{j}$$



$$24. \oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$$

$$= \mu_0 (i + 3i - 6i + 7i)$$

$$= 5\mu_0 i = 3.77 \times 10^{-8} \text{ T} \cdot \text{m}$$



0 wire

$$26. \oint \vec{B} \cdot d\vec{s} = B_{p, \text{pipe}} \cdot 2\pi \cdot 2R = \mu_0 i$$

$$\therefore B_{p, \text{pipe}} = \frac{\mu_0 i}{4\pi R}$$

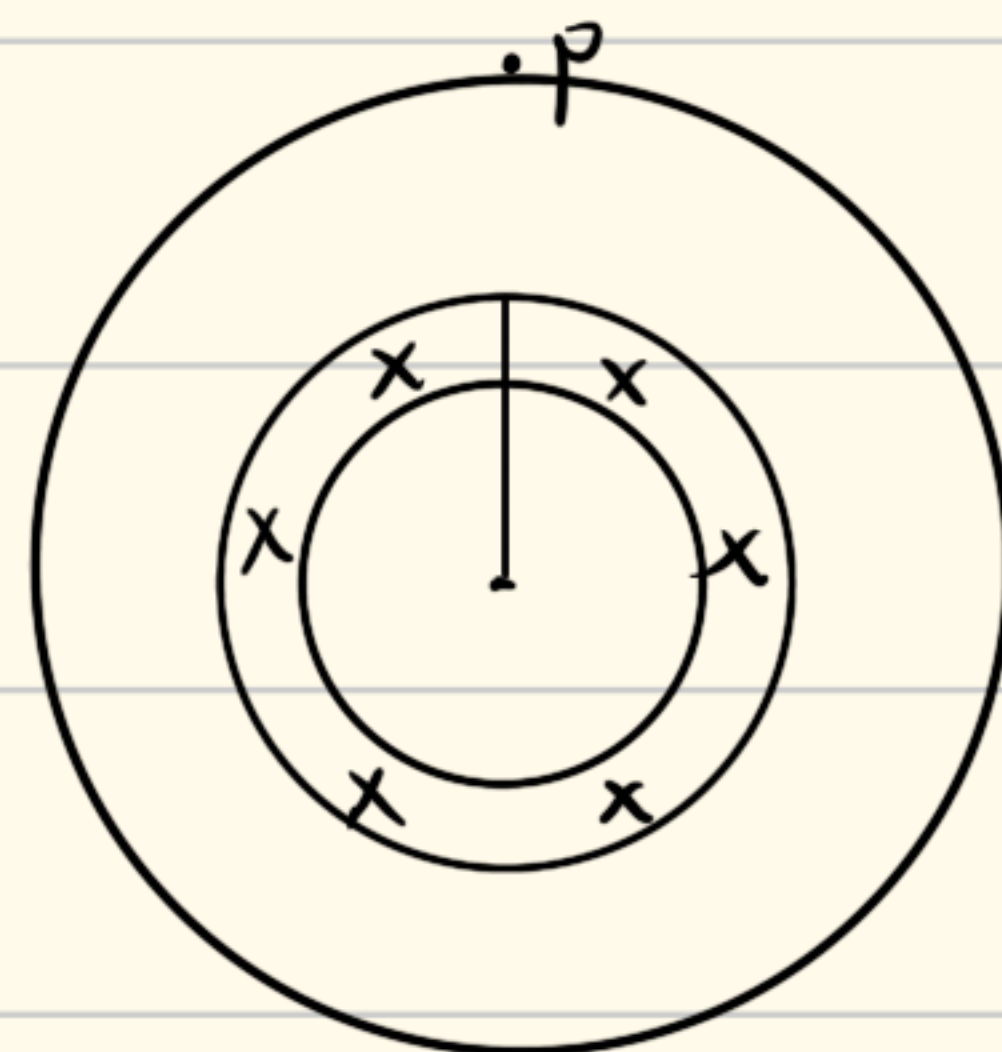
$$\therefore B_{p, \text{wire}} = \frac{\mu_0 i_w}{2\pi R}$$

$$B_{\text{cen, pipe}} = 0, B_{\text{cen, wire}} = \frac{\mu_0 i_w}{2\pi \cdot 3R}$$

$$\therefore \vec{B}_{p, \text{pipe}} + \vec{B}_{p, \text{wire}} = \vec{B}_{\text{cen, wire}}$$

$$\therefore i_w = \frac{3}{8} i = 0.795 \text{ mA}$$

direction: into the page



$$27. (a) B_s = \mu_0 i_0 N = 4.40 \times 10^{-5} \text{ T}$$

$$B_c = \frac{\mu_0 i}{2\pi d} = 1 \times 10^{-6} / d \text{ T}$$

$$\tan 45^\circ = \frac{B_c}{B_s} \quad \therefore d = 2.27 \text{ cm}$$

$$(b) B = \sqrt{B_s^2 + B_c^2} = 6.22 \times 10^{-5} \text{ T}$$

$$44. \oint \vec{B}_1 \cdot d\vec{s} = B_1 \cdot 2\pi R = \mu_0 i_1$$

$$\therefore B_1 = 8 \times 10^{-8} \text{ T}$$

$$\vec{B}_1 = (8.00 \times 10^{-8} \text{ T}) \hat{j}$$

$$\oint \vec{B}_2 \cdot d\vec{s} = B_2 \cdot 2\pi R = \mu_0 i_2$$

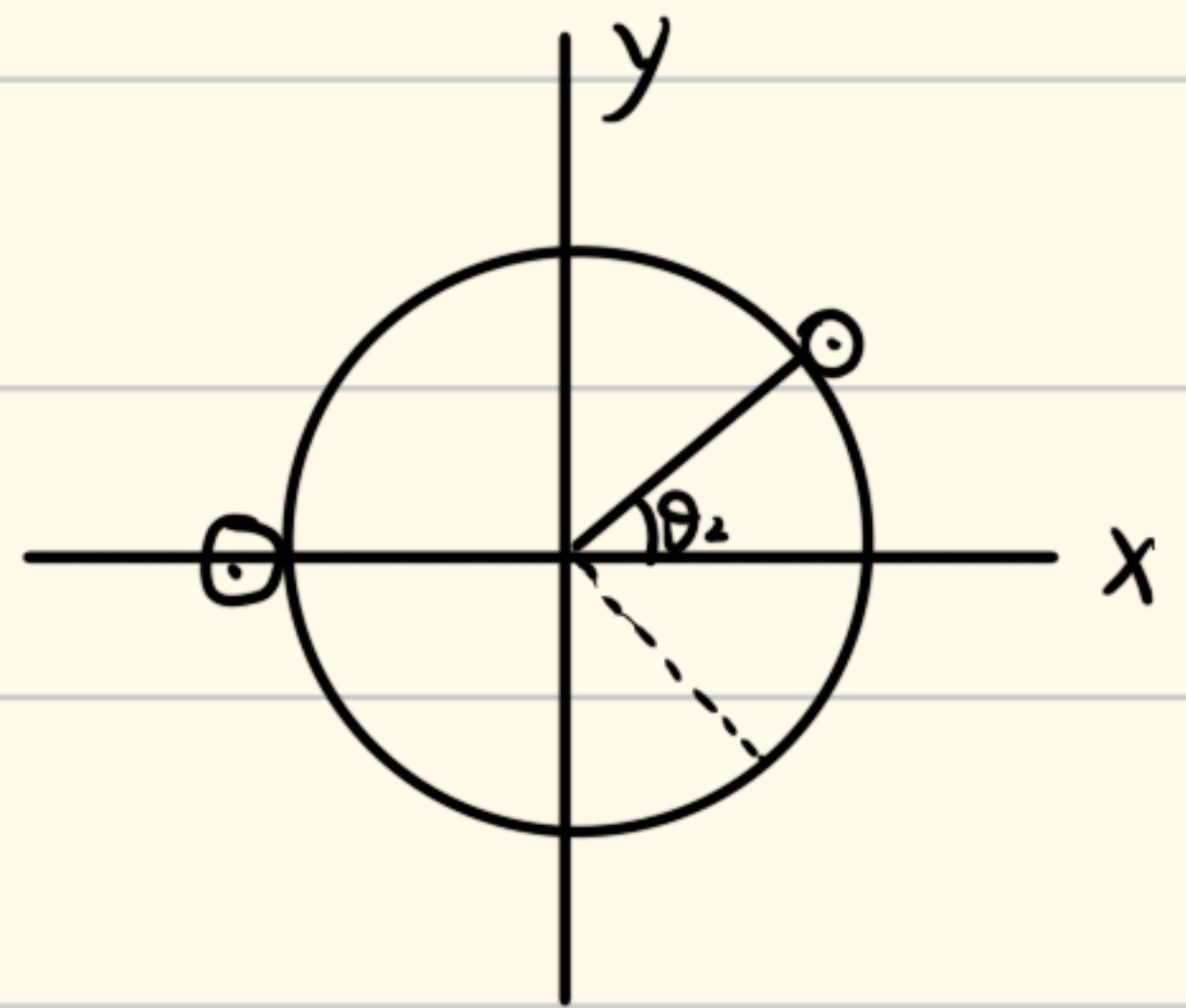
$$\therefore B_2 = 5.33 \times 10^{-8} \text{ T}$$

$$\begin{aligned} \therefore \vec{B}_2 &= B_2 \cos(\theta_2 - \frac{\pi}{2}) \hat{i} + B_2 \sin(\theta_2 - \frac{\pi}{2}) \hat{j} \\ &= B_2 \sin \theta_2 \hat{i} - B_2 \cos \theta_2 \hat{j} \end{aligned}$$

$$\therefore \vec{B} = \vec{B}_1 + \vec{B}_2 = B_2 \sin \theta_2 \hat{i} + (B_1 - B_2 \cos \theta_2) \hat{j}$$

$$\therefore B = \sqrt{B_2^2 + B_1^2 - 2B_1 B_2 \cos \theta_2}$$

$$\therefore \theta_2 = 70.5^\circ$$



48. Set B_1 as the magnitude of magnetic field produced by the larger semicircle, B_2 produced by the smaller semicircle.

$$B_1 = \int d\vec{B} = \int_0^{\pi R} \frac{\mu_0}{4\pi} \frac{i ds}{r^2} = \frac{\mu_0 i}{4R_1}$$

$$B_2 + B_1 = 47.25 \mu T$$

$$B_2 - B_1 = 15.75 \mu T.$$

$$\therefore B_1 = 15.75 \mu T, B_2 = 31.50 \mu T.$$

$$\therefore B_2 = \frac{\mu_0 i}{4R_2}$$

$$\therefore R_2 = \frac{1}{2} R_1 = 2.25 \text{ cm}.$$

$$53. \vec{B} = \frac{\mu_0 i}{2\pi dk} = (3.25 \times 10^{-6} \text{ T}) \hat{k}$$

$$\vec{F} = e\vec{v} \times \vec{B} = (-1.98 \times 10^{-22}) \hat{i}.$$

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Chapter 30:

7. (a) [Eq. 29-2] : $\vec{B}(z) = \frac{\mu_0}{2\lambda} \frac{\vec{m}}{z^3}$.

$$B = \frac{\mu_0 i \pi R^2}{2\lambda x^3}$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = B \cdot \pi r^2 = \frac{\pi \mu_0 i R^2 r^2}{2x^3}$$

$$(b) \mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d\pi \mu_0 i R^2 r^2}{2x^3} \cdot \frac{1}{dt}$$

$$= -\frac{\pi \mu_0 i R^2 r^2}{2} \cdot \frac{dx^{-3}}{dt}$$

$$= -\frac{\pi \mu_0 i R^2 r^2}{2} \cdot -\frac{3}{x^4} \frac{dx}{dt}$$

$$= \frac{3\pi \mu_0 i R^2 r^2 v}{2x^4}$$

(c) Counterclockwise.

15. $\mathcal{E}_L = -L \frac{di}{dt}$ $L \frac{di}{dt} + iR - \mathcal{E} = 0$

$$i = \frac{\mathcal{E}}{R} (1 - e^{-Rt/L}) = 2.00 \text{ mA}$$

(a) $L = 45.5 \text{ H}$

(b) $U = \frac{1}{2} L i^2 = 9.11 \times 10^{-5} \text{ J}$

17. (a) $\mathcal{E}_1 = -M \frac{di_2}{dt}$

$$M = 1.19 \times 10^{-3} \text{ H}$$

(b) $M = \frac{N_2 \Phi_2}{i_1}$

$$N_2 \Phi_2 = 1.61 \times 10^{-3} \text{ Wb}$$

$$40. (a) B = \frac{\mu_0 i}{2\pi r}$$

$$\begin{aligned}\Phi_B &= \int_{b+vt}^{2b+vt} \frac{\mu_0 i}{2\pi r} \cdot a dr \\ &= \frac{\mu_0 i a}{2\pi} \ln \frac{2b+vt}{b+vt}\end{aligned}$$

$$\text{When } t=0, \Phi_B = \frac{\mu_0 i a}{2\pi} \ln 2 = 2.1 \times 10^{-8} \text{ Wb}$$

$$\begin{aligned}(b) \mathcal{E} &= -\frac{d\Phi_B}{dt} = -\frac{\mu_0 i a}{2\pi} \left(\frac{b+vt}{2b+vt} \cdot \frac{v(2b+vt) - v(b+vt)}{(b+vt)^2} \right) \\ &= -\frac{\mu_0 i a}{2\pi} \left(\frac{-vb}{2b+vt} \cdot \frac{1}{b+vt} \right)\end{aligned}$$

When $t=0$, center at $r=1.5b$,

$$\mathcal{E} = \frac{\mu_0 i a v}{4\pi b} = 6.072 \times 10^{-9} \text{ V}$$

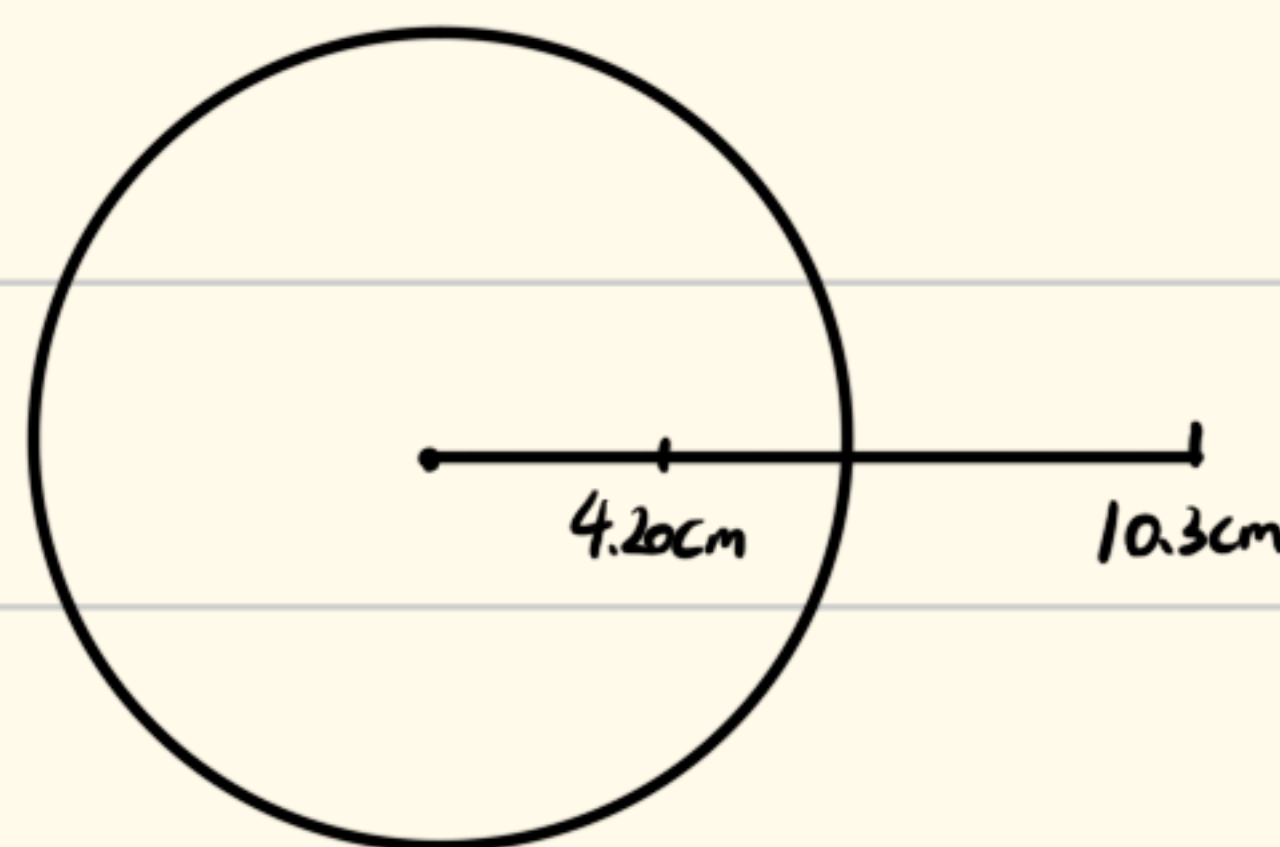
$$i = \frac{\mathcal{E}}{R} = 1.52 \times 10^{-5} \text{ A}$$

$$47. (a) \oint \vec{E} \cdot d\vec{s} = E_1 \cdot 2\pi r_1 = -\frac{d\Phi_{B_1}}{dt} = \pi r_1^2 \frac{dB}{dt}$$

$$\therefore E_1 = 1.37 \times 10^{-4} \text{ V/m}$$

$$(b) \oint \vec{E}_2 \cdot d\vec{s} = E_2 \cdot 2\pi r_2 = -\frac{d\Phi_{B_2}}{dt} = \pi R^2 \frac{dB}{dt}$$

$$\therefore E_2 = 1.14 \times 10^{-4} \text{ V/m}$$



$$53. (a) \mathcal{E}_i = -\frac{d\Phi_B}{dt} = -\frac{\frac{1}{2}a^2 dB}{dt} = -\frac{1}{2}a^2 \cdot (-1.25) = 2.5 \text{ V}$$

$$\mathcal{E}_{\text{net}} = \mathcal{E} + \mathcal{E}_i = 14.5 \text{ V}$$

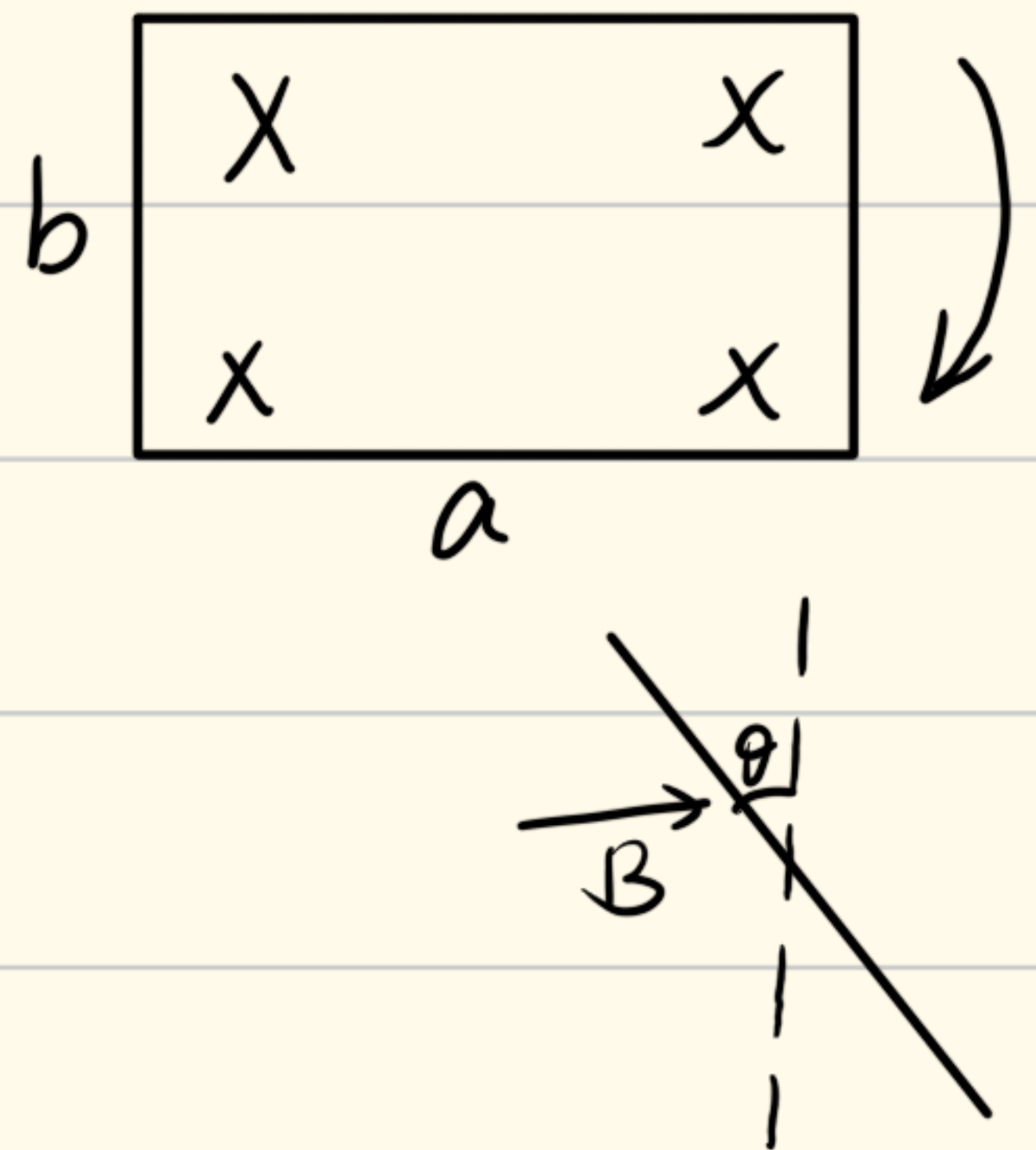
(b) i is counterclockwise

$$63. L_{\text{eq}} = L_1 + \frac{1}{\frac{1}{L_2} + \frac{1}{L_3}} + L_4 = 81.0 \text{ mH}$$

$$\begin{aligned} 69. \mathcal{E} &= -N \frac{d\Phi_B}{dt} \\ &= -NB \frac{d(ab \cos \omega t)}{dt} \\ &= abNB\omega \sin(\omega t) \end{aligned}$$

$$\omega = 2\pi n = 42.6\pi \text{ rad/s}$$

$$\therefore \mathcal{E}_{\text{max}} = abN\omega = 8.63 \times 10^3 \text{ V}$$



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Chapter 31:

3. $L = L_1 + L_2 = 12.0 \text{ mH}$ (in series)

$$C = C_1 + C_2 + C_3 = 14.4 \mu\text{F} \text{ (in parallel)}$$

(a) $\omega_d = \omega = \frac{1}{\sqrt{LC}} = 2.406 \times 10^3 \text{ rad/s}$

$$f = \frac{\omega_d}{2\pi} = 383 \text{ Hz}$$

(b) f remains unchanged.

(c) f decreases (d) f increases (e) f increases

23. (a) The power factor $\cos(42.0^\circ) = 0.74$

(b) leads

(c) $\phi = -42.0^\circ$

$$\tan \phi = \frac{X_L - X_C}{R} < 0.$$

$\therefore X_L < X_C$, largely capacitive.

(d) No. (e) Yes (more capacitive)

(f) No (g) Yes

(h) $P_{\text{avg}} = \varepsilon_{\text{rms}} I_{\text{rms}} \cos(-42.0^\circ) = \frac{\varepsilon_m I_m}{2} \cos(-42.0^\circ) = 21.2 \text{ W}$

(i) All these questions are related to ϕ and avg.

$$26. C_{eq} = \frac{1}{\frac{1}{C} + \frac{1}{C}} = \frac{1}{2}C$$

$$\phi_1 = -25.0^\circ, \phi_2 = 20.0^\circ$$

$$\tan \phi_1 = \frac{X_L - X_{Ceq}}{R}$$

$$X_{Ceq} = \frac{1}{\omega_d C} \quad X_C = \frac{1}{\omega_d C}$$

$$\tan \phi_2 = \frac{X_L - X_C}{R}$$

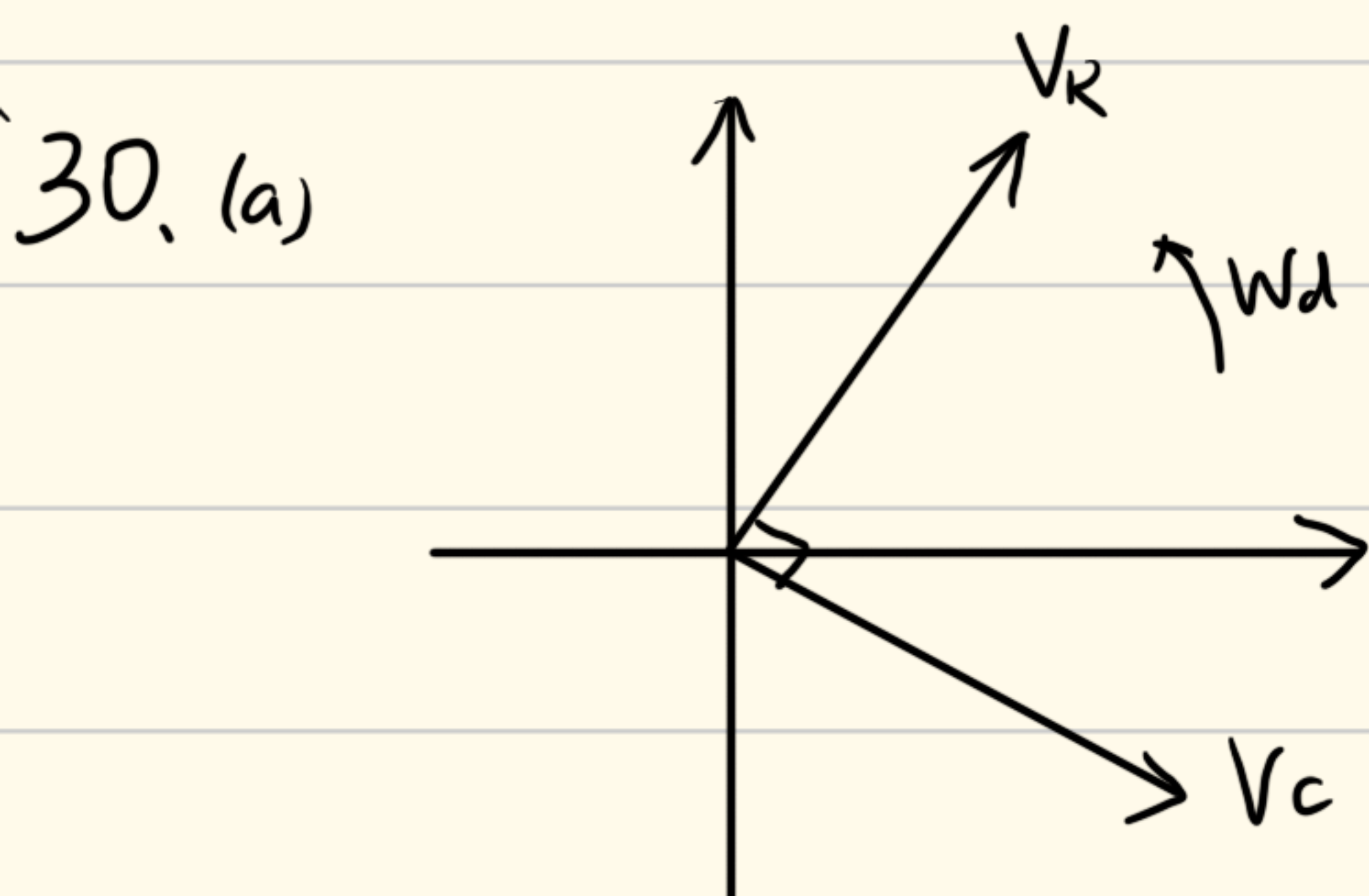
$$\therefore X_{Ceq} = 2X_C$$

$$I = \frac{\epsilon_m}{X_L - X_C}$$

$$(a) R = 73.8 \Omega$$

$$(b) C = \frac{1}{\omega_d X_C} = \frac{1}{2\pi f X_C} = 4.33 \times 10^{-5} F$$

$$(c) L = \frac{X_L}{\omega_d} = \frac{X_L}{2\pi f} = 0.234 H$$



$$(b) \therefore V_R = V_C \quad \therefore IR = IX_C = I \frac{1}{\omega_d C}$$

$$\therefore \omega_d = \frac{1}{CR}$$

$$f_d = \frac{\omega_d}{2\pi} = 113.7 \text{ Hz}$$

$$(c) \tan \phi = \frac{0 - X_C}{R} = -\frac{1}{\omega_d CR} = -\frac{1}{2\pi f_d CR} = -1.00$$

$$\therefore \phi = -45^\circ$$

$$(d) \omega_d = 2\pi f_d = 714.4 \text{ rad/s}$$

$$(e) I = \frac{\epsilon_m}{\sqrt{R^2 + X_C^2}} = 0.170 A$$

$$43. (a) U = U_C + U_L = \frac{q^2}{2C} + \frac{1}{2} L i^2 = 3.56 \times 10^{-6} \text{ J}$$

$$(b) U = \frac{Q^2}{2C}$$

$$Q = \sqrt{2CU} = 4.54 \times 10^{-6} \text{ C}$$

$$(c) U = \frac{1}{2} L I^2$$

$$I = \sqrt{\frac{2U}{L}} = 1.69 \times 10^{-2} \text{ A}$$

$$(d) q = Q \cos \phi$$

$$\phi = \cos^{-1}\left(\frac{q}{Q}\right) = 33.2^\circ$$

$$\text{charging: } \phi = -33.2^\circ$$

$$(e) \text{discharging: } \phi = 33.2^\circ$$

$$49. (a) f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} = 122.6 \text{ Hz}$$

$$(b) Q = C\varepsilon = 1.06 \times 10^{-3} \text{ C}$$

$$i = \omega Q = 2\pi f Q = 0.816 \text{ A}$$

$$58. (a) \text{ When } \omega_d = 2.5 \times 10^4 \text{ rad/s, the circuit is in resonance.}$$

$$X_L = X_C \quad \therefore \omega_d L = \frac{1}{\omega_d C}$$

$$C = \frac{1}{\omega_d^2 L} = 3.56 \times 10^{-6} \text{ F}$$

$$(b) I_s = \frac{\varepsilon_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\therefore R = 1.50 \Omega$$

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Chapter 32:

$$1. (a) \frac{\Phi_{E1}}{\Phi_E} = \frac{\pi r_1^2}{\pi R^2} = \left(\frac{r_1}{R}\right)^2$$

$$\oint \vec{B}_1 \cdot d\vec{s} = B_1 \cdot 2\pi r_1 = \mu_0 \epsilon_0 \frac{d\Phi_{E1}}{dt}$$

$$\therefore B_1 = \frac{\mu_0 \epsilon_0 \frac{d\Phi_{E1}}{dt}}{2\pi r_1} = 6.64 \times 10^{-20} \text{ T}$$

$$(b) \oint \vec{B}_2 \cdot d\vec{s} = B_2 \cdot 2\pi r_2 = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\therefore B_2 = \frac{\mu_0 \epsilon_0 \frac{d\Phi_E}{dt}}{2\pi r_2} = 1.06 \times 10^{-19} \text{ T}$$

$$(c) r \leq R: B \cdot 2\pi r = \mu_0 \epsilon_0 \frac{r^2}{R^2} \frac{d\Phi_E}{dt}$$

$$\therefore B = \frac{\mu_0 \epsilon_0 r}{2\pi R^2} \frac{d\Phi_E}{dt}, \text{ reaches its peak at } r=R,$$

$$B_{\max} = 1.33 \times 10^{-19} \text{ T}$$

$$r \geq R: B \cdot 2\pi r = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

$$\therefore B = \frac{\mu_0 \epsilon_0}{2\pi r} \frac{d\Phi_E}{dt}, \text{ reaches its peak at } r=R.$$

$$\therefore \text{When } r=4.00 \text{ cm, } B_{\max} = 1.33 \times 10^{-19} \text{ T.}$$

$$45. (a) J = \frac{I}{A} = 2.00 \times 10^8 \text{ A/m}^2$$

$$E = \rho J = 0.324 \text{ V/m}$$

$$(b) i_d = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 \rho \frac{di}{dt} = 2.87 \times 10^{-18} \text{ A}$$

$$(c) B_{id} \cdot 2\pi r = \mu_0 i_d \quad \therefore B_{id} = \frac{\mu_0 i_d}{2\pi r}$$

$$B_i \cdot 2\pi r = \mu_0 i \quad B_i = \frac{\mu_0 i}{2\pi r}$$

$$\therefore B_{id}/B_i = i_d/i = 5.31 \times 10^{-18}$$

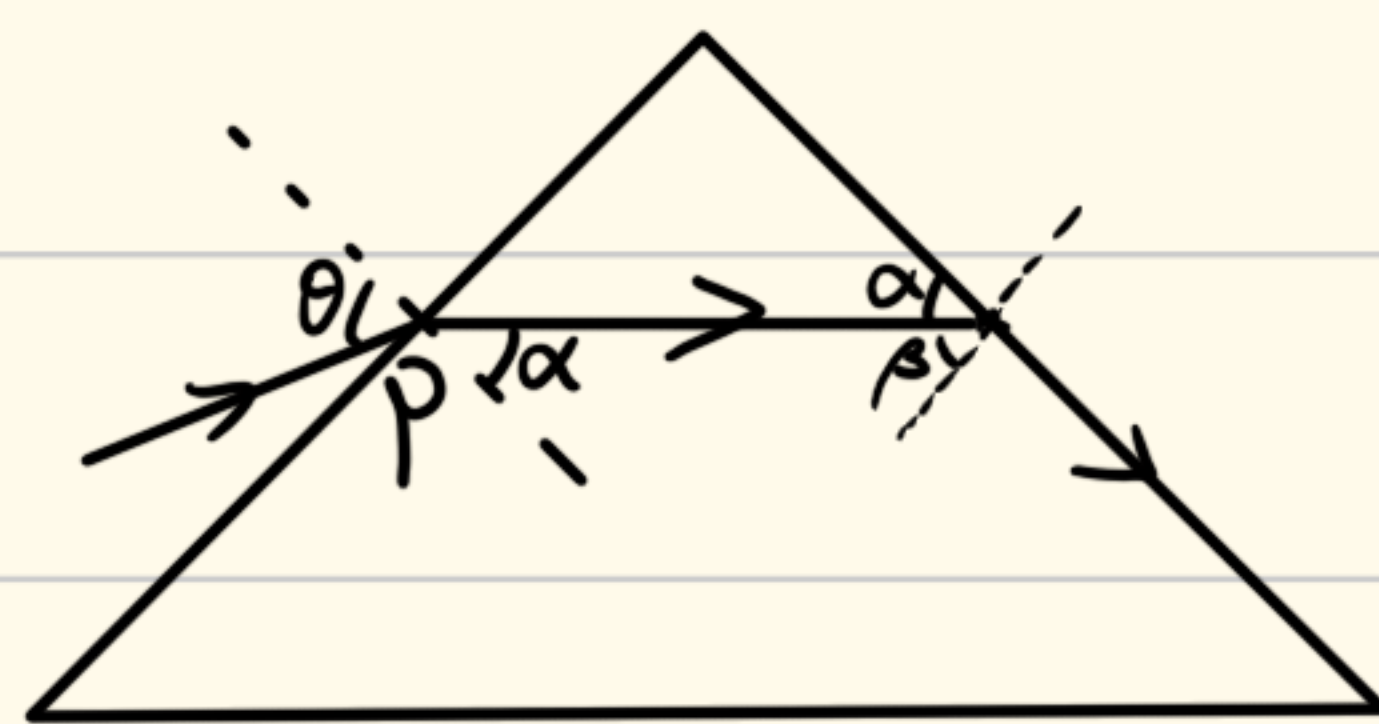
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Chapter 33:

$$7. (a) n = \frac{\sin \theta}{\sin \alpha} = \frac{1}{\sin(90^\circ - \alpha)}$$

$$\therefore \sin \alpha = \frac{\sin \theta}{\sqrt{\sin^2 \theta + 1}}, \cos \alpha = \frac{1}{\sqrt{\sin^2 \theta + 1}}$$

$$n = \sqrt{\sin^2 \theta + 1}$$



$$(b) n_{\max} = \sqrt{2}$$

(c) $\theta \uparrow$, $\alpha \uparrow$, $\beta \downarrow$, the light emerges at Q.

(d) $\theta \downarrow$, $\alpha \downarrow$, $\beta \uparrow$, the light doesn't emerge at Q

$$(e) n = \sqrt{\sin^2 \theta + 1} = 1.30$$

$$\therefore \theta = 56.2^\circ$$

$$8. I_1 = \frac{1}{2} I_0$$

$$I_2 = I_1 \cos^2(\theta_2 - \theta_1)$$

$$I_3 = I_2 \cos^2(\theta_3 - \theta_2)$$

When $\theta_2 = 90^\circ$, it indicates $\cos(\theta_2 - \theta_1) = 0$.

When $\theta_3 = 90^\circ$, it indicates $\cos(\theta_3 - \theta_2) = 0$.

$$\therefore \theta_1 = 0^\circ, \theta_3 = 90^\circ$$

$$\therefore I_{\text{trans}} = \frac{1}{2} I_0 \cos^2(\theta_2 - 0^\circ) \cos^2(90^\circ - \theta_2) = 0.110 I_0$$

$$18. (a) n_1 = \frac{\sin \theta}{\sin \alpha_1} \quad \therefore \sin \alpha_1 = 0.681$$

$$\beta_1 = 90^\circ - \alpha_1$$

$$\sin \beta_1 = \cos \alpha_1 = 0.732$$

$$n_1 = \frac{\sin r_1}{\sin \beta_1} \quad \therefore r_1 = 77.10^\circ$$

$$n_2 = \frac{\sin \theta}{\sin \alpha_2} \quad \sin \alpha_2 = 0.675$$

$$\sin \beta_2 = \cos \alpha_2 = 0.738$$

$$n_2 = \frac{\sin r_2}{\sin \beta_2} \quad \therefore r_2 = 82.34^\circ$$

$$\therefore \Delta r = r_2 - r_1 = 5.24^\circ$$

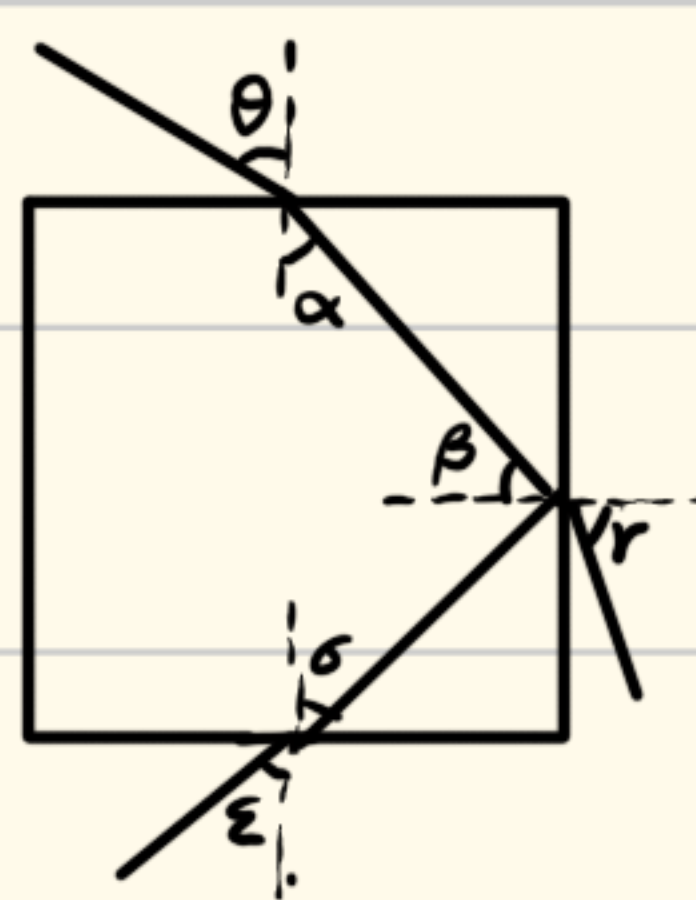
$$(b) \sin \sigma_1 = \cos \beta_1$$

$$n_1 = \frac{\sin \epsilon_1}{\sin \sigma_1} \quad \epsilon_1 = 65.0^\circ$$

$$\sin \sigma_2 = \cos \beta_2$$

$$n_2 = \frac{\sin \epsilon_2}{\sin \sigma_2} \quad \epsilon_2 = 65.0^\circ$$

$$\therefore \Delta \epsilon = \epsilon_2 - \epsilon_1 = 0^\circ$$



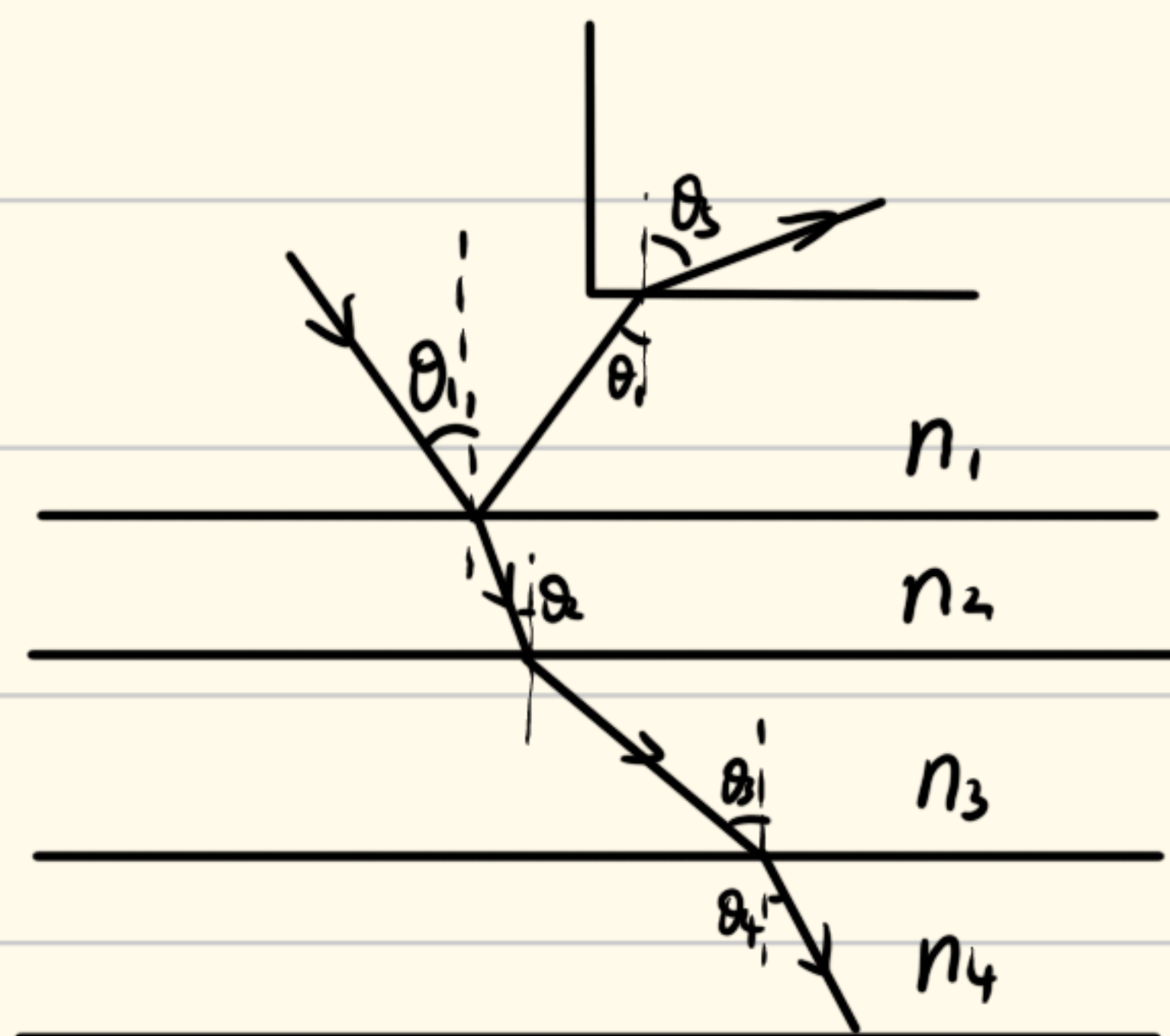
$$23. n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3 = n_4 \sin \theta_4$$

$$\therefore \sin \theta_4 = \frac{n_1 \sin \theta_1}{n_4}$$

$$n_1 \sin \theta_1 = \sin \theta_5$$

$$(a) \theta_5 = \sin^{-1}(n_1 \sin \theta_1) = 56.9^\circ$$

$$(b) \theta_4 = \sin^{-1}\left(\frac{n_1 \sin \theta_1}{n_4}\right) = 28.6^\circ$$



$$47. \theta = \tan^{-1}\left(\frac{n_g}{n_w}\right) = 52.1^\circ$$

$$59. (a) I = \frac{1}{c\mu_0} E_{rms}^2 = \frac{E_0^2}{2c\mu_0}$$

$$\therefore E_0 = \sqrt{2Ic\mu_0} = 0.145 \text{ V/m}$$

$$(b) B_0 = \frac{E_0}{c} = 4.84 \times 10^{-10} \text{ T}$$

$$(c) P = 2\pi r^2 I = 1.76 \times 10^4 \text{ W}$$

$$66. f = \frac{w}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

$$\lambda = \frac{c}{f} = 5.19 \text{ m}$$

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Chapter 35:

$$32. \therefore 2L = m \frac{\lambda}{n_2}, \quad m=0, 1, 2, \dots, 8.$$

$$\therefore L = \frac{8\lambda}{2n_2} = 2800 \text{ nm}.$$

$$L' = L + \Delta L = 3500 \text{ nm}$$

$$m' = \frac{2n_2 L'}{\lambda} = 10.$$

$\therefore$ There'll be $10 + 1 = 11$ ^(m=0) dark fringes.

$$46. 2L = (m + \frac{1}{2}) \frac{\lambda_1}{n}$$

$$n = \frac{(m + \frac{1}{2}) \lambda_1}{2L}$$

Assume $n > 1$, set $m=1$.

$$n = 1.60.$$

$$2L = m \frac{\lambda_2}{n}$$

$$\therefore \lambda_2 = \frac{2nL}{m}$$

$$\text{Visible: } m=2, \lambda_2 = 450.6 \text{ nm}$$

$$56. (a) \sin \theta = \frac{y}{\sqrt{y^2 + D^2}} = 0.0512$$

$$\phi = \frac{2\pi d \sin \theta}{\lambda} = 2.23 \text{ rad}$$

$\therefore P$ is between the maximum (0) and the minimum (π rad).

$$(b) I_p = 4I_0 \cos^2 \frac{1}{2} \phi = 0.784 I_0$$

$$\therefore I_p / I_0 = 0.784$$

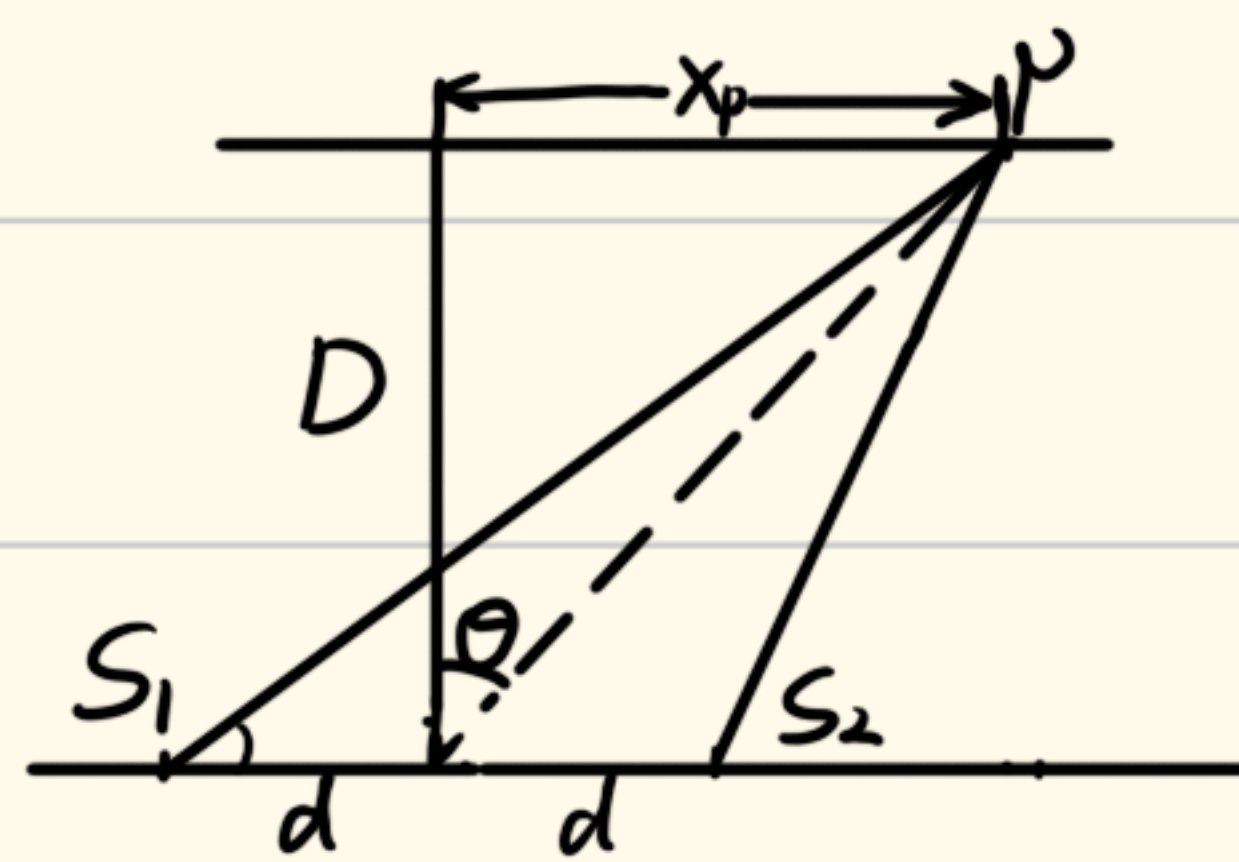
$$63. 2L = (m + \frac{1}{2}) \frac{\lambda}{n}$$

$$(a) m=0, L_0 = \frac{\lambda}{4n} = 79.7 \text{ nm}$$

$$(b) m=1, L_1 = \frac{3\lambda}{4n} = 239 \text{ nm}$$

$$68. (a) OPD = \sqrt{(x_p + d)^2 + D^2} - \sqrt{(x_p - d)^2 + D^2}$$

$$\phi = \frac{OPD}{\lambda} \cdot 2\pi$$



$\therefore$ When $x_p = 0$, ϕ is the minimum 0.

$$(b) m=0$$

$$(c) \phi = 2\pi \cdot \frac{OPD}{\lambda} = \frac{2\pi}{\lambda} (\sqrt{(x_p + d)^2 + D^2} - \sqrt{(x_p - d)^2 + D^2})$$

$$\therefore x_p \rightarrow \infty, \phi_{\max} = \frac{4\pi d}{\lambda} = 12\pi$$

$$(d) m \rightarrow \infty$$

$$(e) \phi = \frac{4\pi d}{\lambda} \frac{x_p}{\sqrt{x_p^2 + D^2}} = 3.71\pi$$

$$(f) I = 4I_0 \cos^2 \frac{1}{2} \phi = 3.23 I_0$$

$\therefore$ intermediate but closer to maximum.

$$74. (a) \theta = \frac{\lambda}{d} = \frac{1}{100} \text{ rad}$$

$$(b) y = L\theta = 0.900 \text{ cm} = 9.00 \text{ mm}$$

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Chapter 36:

17. (a) $a = \frac{\lambda}{\sin \theta} = 7.11 \times 10^3 \text{ nm}$

(b) The fourth bright fringe is missing,
 $d = 4a = 2.84 \times 10^4 \text{ nm}$

(c) $I(\theta) = I_m \left(\frac{\sin\left(\frac{\pi a}{\lambda} \sin \theta\right)}{\frac{\pi a}{\lambda} \sin \theta} \right)^2$

$m=1, \theta = 1.25^\circ, I = 5.67 \text{ mW/cm}^2$

$m=2, \theta = 2.5^\circ, I = 2.83 \text{ mW/cm}^2$

22. (a) $\alpha = \frac{\pi a}{\lambda} \sin \theta$

$k = \frac{\pi a}{\lambda} = 24 \text{ rad}$

$a = 4.66 \times 10^3 \text{ nm}$

(b) $\sin \theta = \frac{m\lambda}{a}$

$\therefore m$ is allowed to be $\pm 1, \pm 2, \dots, \pm 7$.

$\therefore 14$ diffraction minima.

(c) $\theta_{\min} = \sin^{-1} \frac{\lambda}{a} = 7.52^\circ$

(d) $\theta_{\max} = \sin^{-1} \frac{7\lambda}{a} = 66.4^\circ$

$$37. a \ll d$$

$$\sin \theta_1 = \frac{\lambda}{a}$$

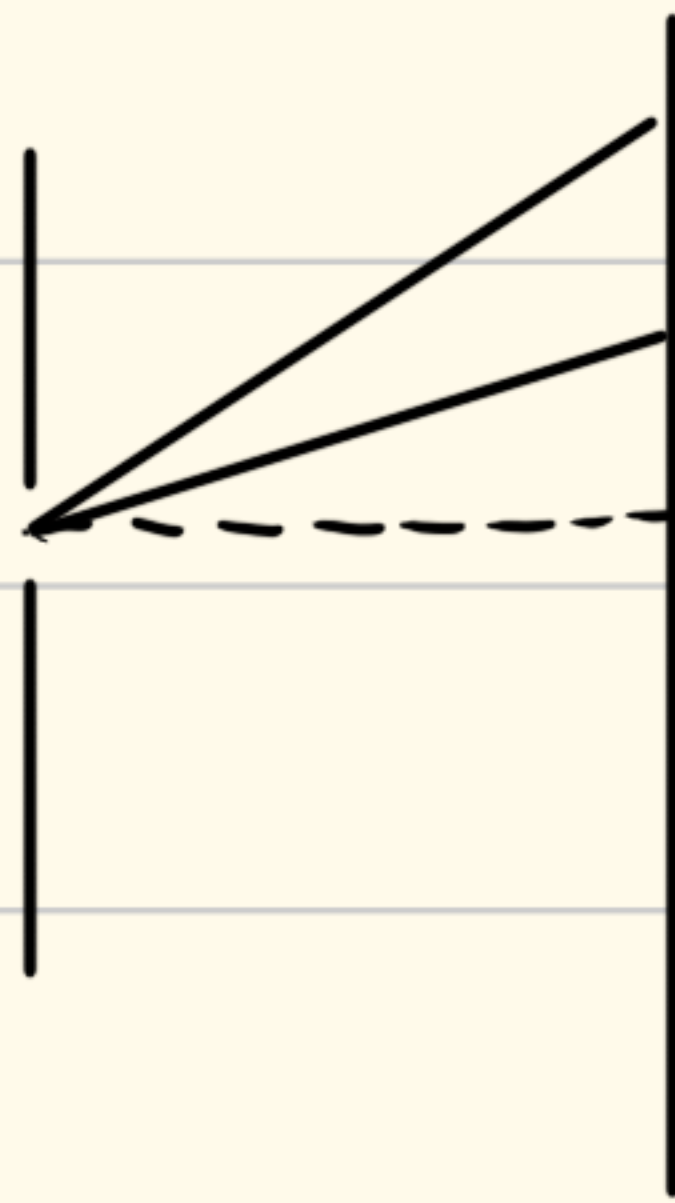
$$\sin \theta_3 = \frac{3\lambda}{a}$$

$$\tan \theta_1 = \frac{y_1}{d}$$

$$\tan \theta_2 = \frac{y_2}{d}$$

$$\sin \theta_1 \sim \tan \theta_1$$

$$\sin \theta_2 \sim \tan \theta_2$$

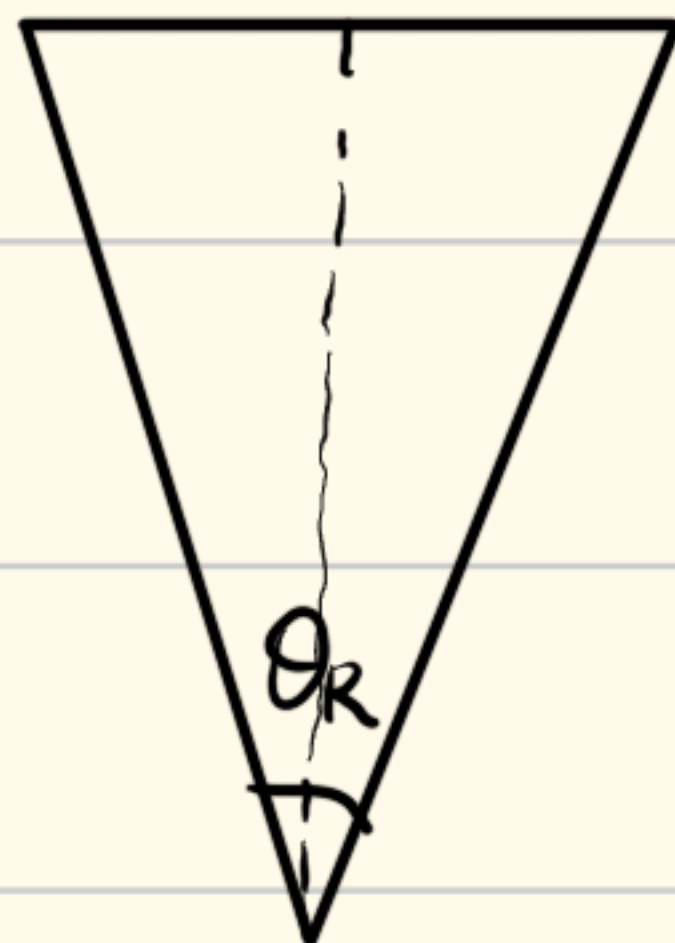


$$\therefore \Delta y = y_2 - y_1 = d \frac{2\lambda}{a} = 3.9 \text{ mm}$$

$$41. (a) \theta_R = 1.22 \frac{\lambda}{a} = 1.49 \times 10^{-4} \text{ rad}$$

$$(b) \tan \frac{\theta_R}{2} = \frac{\frac{L}{2}}{D}$$

$$\therefore D = 9389 \text{ m}$$



$$42. \sin \theta = \frac{\lambda}{a} = \frac{1}{2}$$

$$a : \lambda = 2 : 1$$

$$56. \sin \frac{\theta}{2} = \frac{1.22 \lambda}{d}$$

$$\therefore d = 3.84 \times 10^4 \text{ nm}$$

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Chapter 37:

3. (a) $\Delta X' = \gamma(\Delta X - v\Delta t) = 0$

$$\Delta X = X_2 - X_1 = -500\text{m}$$

$$\therefore V = -1.00 \times 10^8 \text{ m/s}$$

(b) $V < 0$, S' is moving in the negative direction.

$$(c) \Delta t' = \gamma(t - vx/c^2) = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}} = 4.71 \times 10^{-6} \text{ s} > 0.$$

$\therefore$ The big flash comes first.

$$(d) \Delta t' = 4.71 \times 10^{-6} \text{ s} = 4.71 \mu\text{s}.$$

10. (a) $V = \frac{|\Delta\lambda|}{\lambda_0} c = 7.01 \times 10^6 \text{ m/s}.$

(b) $\lambda > \lambda_0$, the relative motion is away from us.

(c) A red shift.

24. $\Delta t' = 54.0 \text{ y}$, $\Delta t_0 = 4.000 \text{ y}.$

$$\Delta t' = \frac{\Delta t_0}{\sqrt{1 - \beta^2}}$$

$$\therefore \beta = 0.9973$$

30. (a) $\vec{v} = 0.620c\hat{i}$, $\vec{u}' = 0.420c\hat{i}$

$$\vec{u} = \frac{\vec{u}' + \vec{v}}{1 + \vec{u}'\vec{v}/c^2} = 0.825c\hat{i}.$$

$$(b) \vec{u} = \vec{u}' + \vec{v} = 1.040c\hat{i}$$

$$(c) \vec{v} = 0.620c \hat{i}, \vec{u}' = -0.420c \hat{i}.$$

$$\vec{u} = \frac{\vec{u}' + \vec{v}}{1 + \vec{u}' \cdot \vec{v} / c^2} = 0.270c \hat{i}$$

$$(d) \vec{u} = \vec{u}' + \vec{v} = 0.200c \hat{i}.$$

$$43. (a) \Delta t_1 = \frac{d}{v} = 26.13 \text{ y}$$

$$(b) \Delta t_2 = \frac{d}{c} = 26.00 \text{ y}$$

$$\Delta t = \Delta t_1 + \Delta t_2 = 52.13 \text{ y}$$

$$(c) \Delta t_1 = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}$$

$$\therefore \Delta t_0 = 2.610 \text{ y}$$

$$56. \Delta x = x_2 - x_1 = 3.0 \times 10^3 \text{ m}, \Delta t = 4.0 \times 10^{-6} \text{ s}, v = 0.65c.$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = 1.316$$

$$(a) t'_1 = \gamma(t - vx_1/c^2) = 0.$$

$$(b) t'_2 = \gamma(t - vx_2/c^2) = -3.3 \mu\text{s}$$

(c) in the reverse sequence.